

A7b-R.R

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```
# Set the base directory for the project
base_dir <- getwd()
setwd(base_dir)
```

```
# Define a helper function to install packages if not already installed
install <- function(pkg) {
  if (!require(pkg, character.only = TRUE)) {
    install.packages(pkg, dependencies = TRUE, quiet = TRUE)
  }
}

# Define a helper function to load packages
load_pkg <- function(pkg) {
  message(paste("Loading package:", pkg))
  library(pkg, character.only = TRUE, quietly = TRUE)
}
```

```
# Required packages for this analysis
pkgs <- c("readxl", "vars", "urca", "tseries", "forecast", "quantmod", "rugarch", "FinTS",
"tsDyn")
```

```
# Apply install and load functions
lapply(pkgs, install)
lapply(pkgs, load_pkg)
```

```
#####
# PART A #
#####

# Downloading data for NVDA ticker from Yahoo Finance
ticker <- "NVDA"
getSymbols(ticker, src = "yahoo", from = Sys.Date() - 5*365, auto.assign = TRUE) data <- get(ticker)

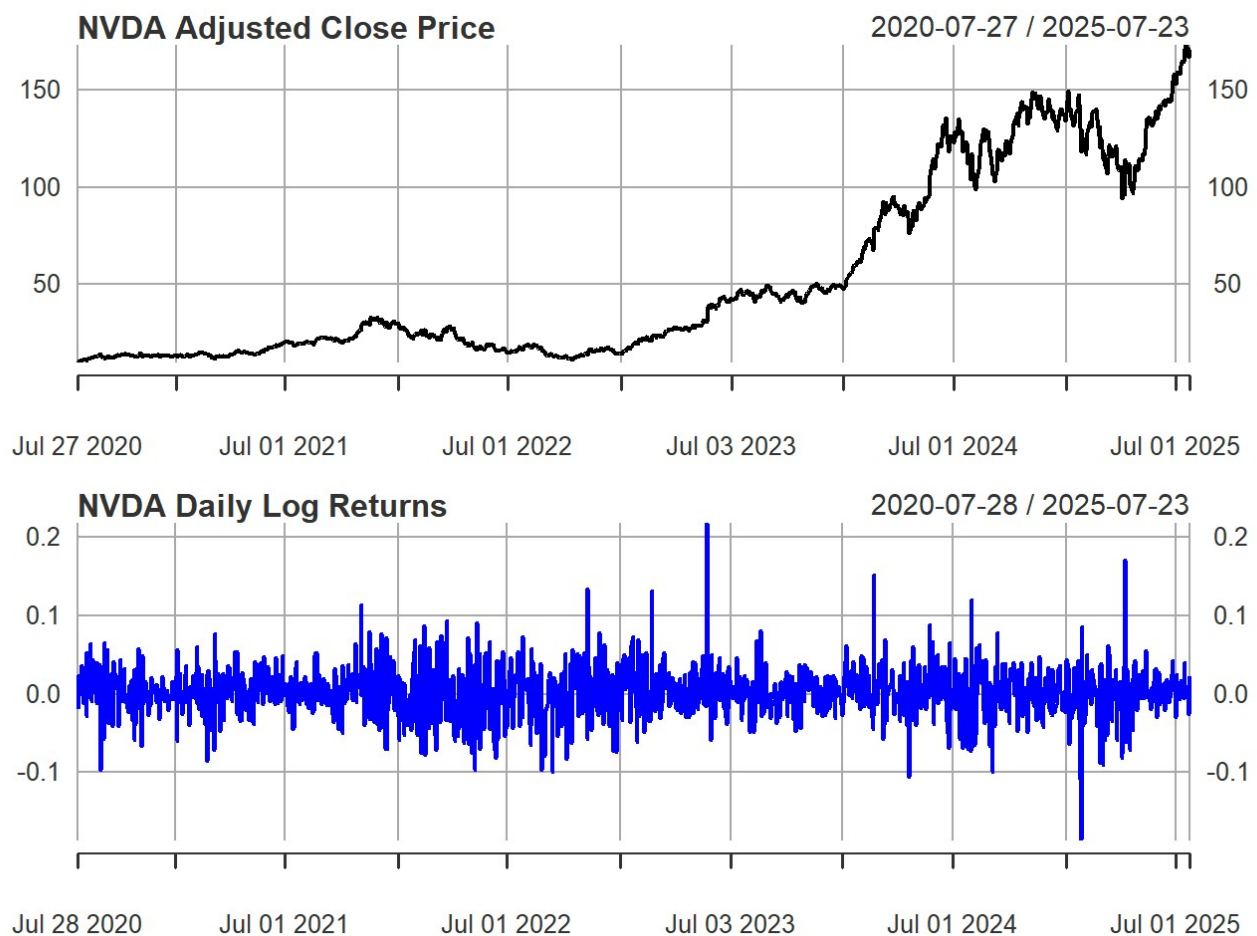
# Use Cl function from quantmod to get the adjusted close series from the data
prices <- Cl(data)

# Compute Log returns
log_returns <- diff(log(prices), lag = 1)
log_returns <- na.omit(log_returns)

# Inspect head
head(log_returns)
```

```
##          NVDA.Close
## 2020-07-28 -1.996484e-02
## 2020-07-29  2.417796e-02
## 2020-07-30  1.408979e-02
## 2020-07-31  7.062008e-05
## 2020-08-03  3.658213e-02
## 2020-08-04  1.956171e-02
```

```
# Plot closing prices and log returns
par(mfrow = c(2,1))
plot(prices, main = paste(ticker, "Adjusted Close Price"))
plot(log_returns, main = paste(ticker, "Daily Log Returns"), col = "blue")
```



```
# ARCH test
ArchTest(ts(log_returns), lags = 5)
```

```
##
## ARCH LM-test; Null hypothesis: no ARCH effects
##
## data: ts(log_returns)
## Chi-squared = 10.618, df = 5, p-value = 0.05951
```

```
# Scale returns by 100 for better convergence
scaled_returns <- log_returns * 100

# Define ARCH(1) specification
spec_arch <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1, 0)),
  mean.model = list(armaOrder = c(0, 0), include.mean = TRUE),
  distribution.model = "norm"
)

# Fit ARCH(1) model
fit_arch <- ugarchfit(spec = spec_arch, data = scaled_returns)
show(fit_arch)
```

```

##
## *-----*
## *          GARCH Model Fit          *
## *-----*
##
## Conditional Variance Dynamics
## -----
## GARCH Model   : sGARCH(1,0)
## Mean Model    : ARFIMA(0,0,0)
## Distribution   : norm
##
## Optimal Parameters
## -----
##           Estimate Std. Error  t value Pr(>|t|)
## mu         0.241123   0.092991   2.5930 0.009515
## omega      10.351439   0.499371  20.7289 0.000000
## alpha1     0.054279   0.030583   1.7748 0.075925
##
## Robust Standard Errors:
##           Estimate Std. Error  t value Pr(>|t|)
## mu         0.241123   0.092938   2.5944 0.009475
## omega      10.351439   0.990689  10.4487 0.000000
## alpha1     0.054279   0.040143   1.3521 0.176331
##
## LogLikelihood : -3273.593
##
## Information Criteria
## -----
##
## Akaike          5.2300
## Bayes           5.2423
## Shibata         5.2300
## Hannan-Quinn    5.2346
##
## Weighted Ljung-Box Test on Standardized Residuals
## -----
##                               statistic p-value
## Lag[1]                      1.759  0.1848
## Lag[2*(p+q)+(p+q)-1][2]     2.020  0.2595
## Lag[4*(p+q)+(p+q)-1][5]     4.685  0.1801
## d.o.f=0
## H0 : No serial correlation
##
## Weighted Ljung-Box Test on Standardized Squared Residuals
## -----
##                               statistic p-value
## Lag[1]                      0.007216  0.9323
## Lag[2*(p+q)+(p+q)-1][2]     0.977332  0.5056
## Lag[4*(p+q)+(p+q)-1][5]     4.739243  0.1752
## d.o.f=1
##
## Weighted ARCH LM Tests
## -----
##           Statistic Shape Scale P-Value

```

```

## ARCH Lag[2]      1.934 0.500 2.000 0.16432
## ARCH Lag[4]      5.619 1.397 1.611 0.06215
## ARCH Lag[6]      7.564 2.222 1.500 0.05235
##
## Nyblom stability test
## -----
## Joint Statistic:  0.5609
## Individual Statistics:
## mu      0.1096
## omega   0.3066
## alpha1  0.1428
##
## Asymptotic Critical Values (10% 5% 1%)
## Joint Statistic:      0.846 1.01 1.35
## Individual Statistic:  0.35 0.47 0.75
##
## Sign Bias Test
## -----
##              t-value   prob sig
## Sign Bias      0.4285 0.6684
## Negative Sign Bias 0.8490 0.3961
## Positive Sign Bias 0.5853 0.5585
## Joint Effect    4.1766 0.2430
##
##
## Adjusted Pearson Goodness-of-Fit Test:
## -----
##   group statistic p-value(g-1)
## 1    20      49.49   0.0001557
## 2    30      65.78   0.0001120
## 3    40      69.94   0.0017055
## 4    50      77.93   0.0053378
##
##
## Elapsed time : 0.2281311

```

```

# Define GARCH(1,1) specification
spec_garch <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1, 1)),
  mean.model = list(armaOrder = c(0, 0), include.mean = TRUE),
  distribution.model = "norm"
)

# Fit GARCH(1,1) model
fit_garch <- ugarchfit(spec = spec_garch, data = scaled_returns)
show(fit_garch)

```

```

##
## *-----*
## *          GARCH Model Fit          *
## *-----*
##
## Conditional Variance Dynamics
## -----
## GARCH Model   : sGARCH(1,1)
## Mean Model    : ARFIMA(0,0,0)
## Distribution   : norm
##
## Optimal Parameters
## -----
##          Estimate  Std. Error  t value Pr(>|t|)
## mu        0.305229   0.088533   3.4476 0.000566
## omega      0.771645   0.296807   2.5998 0.009327
## alpha1     0.078559   0.024766   3.1720 0.001514
## beta1      0.854543   0.044227  19.3217 0.000000
##
## Robust Standard Errors:
##          Estimate  Std. Error  t value Pr(>|t|)
## mu        0.305229   0.093178   3.2758 0.001054
## omega      0.771645   0.483930   1.5945 0.110815
## alpha1     0.078559   0.033181   2.3676 0.017904
## beta1      0.854543   0.057870  14.7667 0.000000
##
## LogLikelihood : -3252.023
##
## Information Criteria
## -----
##
## Akaike          5.1972
## Bayes           5.2136
## Shibata         5.1971
## Hannan-Quinn    5.2033
##
## Weighted Ljung-Box Test on Standardized Residuals
## -----
##
##          statistic p-value
## Lag[1]                0.5521  0.4575
## Lag[2*(p+q)+(p+q)-1][2] 0.6063  0.6457
## Lag[4*(p+q)+(p+q)-1][5] 2.9076  0.4242
## d.o.f=0
## H0 : No serial correlation
##
## Weighted Ljung-Box Test on Standardized Squared Residuals
## -----
##
##          statistic p-value
## Lag[1]                0.2138  0.6438
## Lag[2*(p+q)+(p+q)-1][5] 0.6363  0.9348
## Lag[4*(p+q)+(p+q)-1][9] 0.9488  0.9883
## d.o.f=2
##
## Weighted ARCH LM Tests

```

```
## -----
##           Statistic Shape Scale P-Value
## ARCH Lag[3]    0.4774 0.500 2.000 0.4896
## ARCH Lag[5]    0.7087 1.440 1.667 0.8208
## ARCH Lag[7]    0.7823 2.315 1.543 0.9462
##
## Nyblom stability test
## -----
## Joint Statistic: 0.74
## Individual Statistics:
## mu      0.1630
## omega   0.1180
## alpha1  0.0857
## beta1   0.1203
##
## Asymptotic Critical Values (10% 5% 1%)
## Joint Statistic:      1.07 1.24 1.6
## Individual Statistic: 0.35 0.47 0.75
##
## Sign Bias Test
## -----
##           t-value   prob sig
## Sign Bias      0.59014 0.5552
## Negative Sign Bias 0.09399 0.9251
## Positive Sign Bias 0.96324 0.3356
## Joint Effect    3.69810 0.2960
##
##
## Adjusted Pearson Goodness-of-Fit Test:
## -----
##   group statistic p-value(g-1)
## 1    20      57.15   1.082e-05
## 2    30      67.45   6.705e-05
## 3    40      89.15   8.481e-06
## 4    50      92.85   1.571e-04
##
##
## Elapsed time : 0.1283228
```

```
# Compare model AIC
cat("ARCH(1) AIC:", infocriteria(fit_arch)[1], "\n")
```

```
## ARCH(1) AIC: 5.229996
```

```
cat("GARCH(1,1) AIC:", infocriteria(fit_garch)[1], "\n")
```

```
## GARCH(1,1) AIC: 5.197163
```

```

# Forecast next 90 days
forecast_days <- 90
garch_forecast <- ugarchforecast(fit_garch, n.ahead = forecast_days)
forecast_vol <- sigma(garch_forecast)

# In-sample volatility and its time index
in_sample_vol <- sigma(fit_garch)
in_sample_idx <- index(in_sample_vol)

# Get last 90 days of conditional volatility from the sample
n_hist <- 90
last_hist_idx <- tail(index(in_sample_vol), n_hist)      # 90 in-sample dates
last_hist_vol <- tail(in_sample_vol, n_hist)            # 90 in-sample volatilities

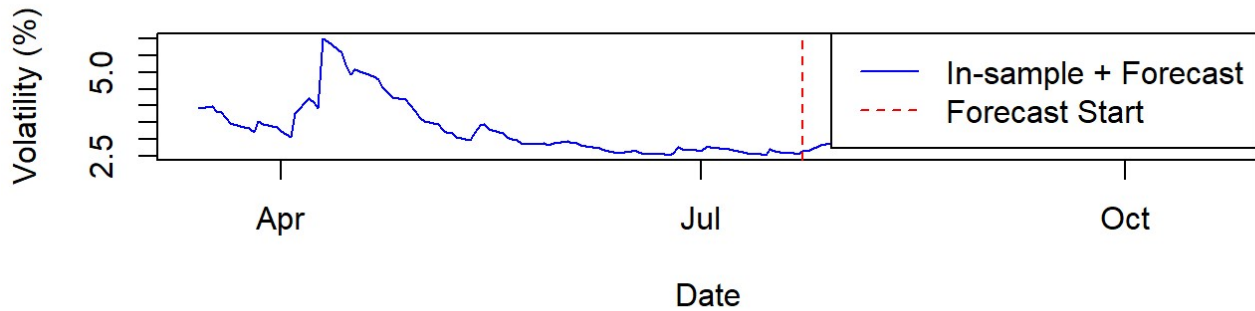
# Get the next 90 days after the last date of the sample
first_forecast_date <- last(last_hist_idx) + 1
forecast_idx <- seq.Date(from = first_forecast_date, by = "day", length.out = forecast_days)

# Combine all dates and all volatilities
all_dates <- c(last_hist_idx, forecast_idx)
all_vol <- c(as.numeric(last_hist_vol), as.numeric(forecast_vol))

# Plot the Conditional Volatility of the last 90 days and the forecast for the next 90 days
plot(all_dates, all_vol, type = "l", col = "blue",
     ylab = "Volatility (%)",
     xlab = "Date",
     main = "Conditional Volatility and 90-Day Forecast")
abline(v = last(last_hist_idx), col = "red", lty = 2)
legend("topright", legend = c("In-sample + Forecast", "Forecast Start"),
     col = c("blue", "red"), lty = c(1,2))

```


Conditional Volatility and 90-Day Forecast



```
#####  
# Part B #  
#####  
  
# File Config  
datasets_dir <- "datasets"  
datasets_path <- file.path(base_dir, datasets_dir)  
file_name <- "CMO-Historical-Data-Monthly.xlsx"  
file_path <- file.path(datasets_path, file_name)  
  
# Read Excel File and handle the Date column  
data_raw <- read_excel(file_path, sheet = "Monthly Prices", skip = 6)  
data_raw <- as.data.frame(data_raw)  
names(data_raw) <- trimws(names(data_raw))  
names(data_raw)[1] <- "Year_Month"  
data_raw$Year_Month <- as.Date(paste0(  
  substr(data_raw$Year_Month, 1, 4), "-",  
  substr(data_raw$Year_Month, 6, 7), "-01"))
```

```

# Clean up undesirable string patterns for as.numeric conversion
clean_numeric_column <- function(vec) {
  vec <- as.character(vec)
  vec <- trimws(vec)
  vec[vec %in% c("", "-", "NA", "n.a.", "N/A", ".", "#N/A N/A")] <- NA
  suppressWarnings(as.numeric(vec))
}

# Data pre-processing per category
prepare_category_df <- function(df, cols) {
  # Subset data frame to only needed columns (assumes df includes Year_Month)
  df_subset <- df[, c("Year_Month", cols), drop=FALSE]

  # Clean columns for numeric conversion
  for (col in cols) {
    df_subset[[col]] <- as.character(df_subset[[col]])
    df_subset[[col]] <- trimws(df_subset[[col]])
    # Replace common non-numeric placeholders with NA
    df_subset[[col]][df_subset[[col]] %in% c("", "-", "NA", "n.a.", "N/A", ".", "#N/A N/A")] <- NA
    # Then convert to numeric
    df_subset[[col]] <- suppressWarnings(as.numeric(df_subset[[col]]))
  }

  # Remove any rows with NA
  n_before <- nrow(df_subset)
  df_subset <- na.omit(df_subset)
  n_after <- nrow(df_subset)
  cat(sprintf("Rows before cleaning: %d. After omitting NA: %d. Removed: %d\n", n_before,
n_after, n_before - n_after))

  # Convert to base data.frame to allow rownames
  df_subset <- as.data.frame(df_subset)

  # Set row names as dates for convenience (optional)
  rownames(df_subset) <- as.character(df_subset$Year_Month)

  # Remove Year_Month column to keep only numeric columns
  df_subset$Year_Month <- NULL

  # Verify all columns numeric and no NAs remain
  for (col in cols) {
    if (!is.numeric(df_subset[[col]])) stop(paste("Column", col, "is not numeric after cleaning!"))
    if (any(is.na(df_subset[[col]]))) stop(paste("Column", col, "still contains NA after cleaning!"))
  }

  # Log transform prices
  log_df <- log(df_subset)

  # Differencing log prices
  log_diff_df <- diff(as.matrix(log_df))
  log_diff_df <- as.data.frame(log_diff_df)

```

```

    return(list(log = log_df, diff_log = log_diff_df))
}

# Augmented Dickey-Fuller test per column for stationarity of differenced log prices
adf_test <- function(diff_df) {
  cat("ADF Test Results (on differenced log prices):\n")
  for (col in colnames(diff_df)) {
    res <- adf.test(diff_df[[col]])
    stationary <- ifelse(res$p.value < 0.05, "Stationary", "Non-stationary")
    cat(sprintf("%-15s: ADF Stat = %.3f, p-value = %.4f --> %s\n", col, res$statistic, res
    $p.value, stationary))
  }
}

# VAR lag selection, fitting, impulse response plotting
run_var <- function(diff_df) {
  lag_sel <- VARselect(diff_df, lag.max = 12, type = "const")
  cat("Lag order selection (AIC, BIC, HQIC):\n")
  print(lag_sel$selection)

  # Extract lag as numeric scalar (important!)
  selected_lag <- as.numeric(lag_sel$selection["AIC(n)"])
  cat("Selected lag order by AIC:", selected_lag, "\n")

  # Fit VAR model with numeric lag
  var_model <- VAR(diff_df, p = selected_lag, type = "const")

  cat("VAR Model summary:\n")
  print("VAR Model object created. Checking its structure:")
  print(summary(var_model))

  # Impulse response function plots
  irf_res <- irf(var_model, n.ahead = 10, ortho = FALSE, boot = FALSE)
  plot(irf_res)

  return(list(model = var_model, lag = selected_lag))
}

# Johansen cointegration test wrapper and estimation
run_johansen <- function(log_df, lag_diff) {
  johansen_result <- ca.jo(log_df, type = "trace", ecdet = "const", K = lag_diff + 1)
  cat("Johansen cointegration test summary:\n")
  print(summary(johansen_result))
  return(johansen_result)
}

get_coint_rank <- function(johansen_result) {
  trace_stats <- johansen_result@teststat
  crit_vals <- johansen_result@cval[, "5pct"]
  estimated_rank <- sum(trace_stats > crit_vals)
  cat(sprintf("Estimated cointegration rank (5%% significance): %d\n", estimated_rank))
  return(estimated_rank)
}

```

```

fit_vecm <- function(log_df, coint_rank, lag_diff) {
  vecm_model <- VECM(log_df, lag = lag_diff, r = coint_rank, estim = "ML")
  cat("VECM fit summary:\n")
  print(summary(vecm_model))
}

# High-Level function for category-wise analysis
do_analysis <- function(df, category_name, cols) {
  cat("\n===== \n")
  cat(sprintf("Analyzing %s Commodities\n", category_name))
  cat("===== \n")

  # Check all columns present
  if (!all(cols %in% colnames(df))) {
    missing_cols <- cols[!cols %in% colnames(df)]
    stop(paste("Error: Missing columns in data frame:", paste(missing_cols, collapse = ",
  )))
  }

  prepared <- prepare_category_df(df, cols)
  log_df <- prepared$log
  diff_log_df <- prepared$diff_log

  cat("\nADF Test for Stationarity:\n")
  adf_test(diff_log_df)

  cat("\nFitting VAR Model:\n")
  var_results <- run_var(diff_log_df)
  selected_lag <- as.numeric(var_results$lag)

  cat("\nConducting Johansen Cointegration Test:\n")
  johansen_res <- run_johansen(log_df, lag_diff = selected_lag)

  rank <- get_coint_rank(johansen_res)
  if (rank > 0) {
    cat("\nFitting VECM Model:\n")
    fit_vecm(log_df, rank, selected_lag)
  } else {
    cat("\nNo significant cointegration detected; skipping VECM.\n")
  }
}

```

```

# Define your analysis groups (edit these lists for your own use case)
crude_oil_cols <- c('CRUDE_BRENT', 'CRUDE_WTI', 'CRUDE_DUBAI')
natural_gas_cols <- c('NGAS_US', 'NGAS_EUR', 'NGAS_JP')
coal_cols <- c('COAL_AUS', 'COAL_SAFRICA')
metals_cols <- c('GOLD', 'SILVER', 'COPPER', 'ALUMINUM')
wheat_cols <- c('WHEAT_US_HRW', 'WHEAT_US_SRW')
sugar_cols <- c('SUGAR_EU', 'SUGAR_US', 'SUGAR_WLD')

do_analysis(data_raw, "Crude Oil", crude_oil_cols)

```

```
##  
## =====  
## Analyzing Crude Oil Commodities  
## =====  
## Rows before cleaning: 786. After omitting NA: 522. Removed: 264  
##  
## ADF Test for Stationarity:  
## ADF Test Results (on differenced log prices):
```

```
## CRUDE_BRENT      : ADF Stat = -8.470, p-value = 0.0100 --> Stationary
```

```
## CRUDE_WTI        : ADF Stat = -8.358, p-value = 0.0100 --> Stationary
```

```
## CRUDE_DUBAI      : ADF Stat = -8.750, p-value = 0.0100 --> Stationary
```

```

## Fitting VAR Model:
## Lag order selection (AIC, BIC, HQIC):
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      6      2      1      6
## Selected lag order by AIC: 6
## VAR Model summary:
## [1] "VAR Model object created. Checking its structure:"
##
## VAR Estimation Results:
## =====
## Endogenous variables: CRUDE_BRENT, CRUDE_WTI, CRUDE_DUBAI
## Deterministic variables: const
## Sample size: 515
## Log Likelihood: 2818.161
## Roots of the characteristic polynomial:
## 0.8112 0.8112 0.7567 0.7567 0.7553 0.7553 0.7315 0.7315 0.6989 0.6989 0.6984 0.6984 0.6
761 0.6761 0.6392 0.6392 0.4962 0.03671
## Call:
## VAR(y = diff_df, p = selected_lag, type = "const")
##
##
## Estimation results for equation CRUDE_BRENT:
## =====
## CRUDE_BRENT = CRUDE_BRENT.l1 + CRUDE_WTI.l1 + CRUDE_DUBAI.l1 + CRUDE_BRENT.l2 + CRUDE_W
TI.l2 + CRUDE_DUBAI.l2 + CRUDE_BRENT.l3 + CRUDE_WTI.l3 + CRUDE_DUBAI.l3 + CRUDE_BRENT.l4 +
CRUDE_WTI.l4 + CRUDE_DUBAI.l4 + CRUDE_BRENT.l5 + CRUDE_WTI.l5 + CRUDE_DUBAI.l5 + CRUDE_BRE
NT.l6 + CRUDE_WTI.l6 + CRUDE_DUBAI.l6 + const
##
##
##           Estimate Std. Error t value Pr(>|t|)
## CRUDE_BRENT.l1 -0.234887   0.193035  -1.217   0.2243
## CRUDE_WTI.l1    0.200712   0.126207   1.590   0.1124
## CRUDE_DUBAI.l1  0.334796   0.167079   2.004   0.0456 *
## CRUDE_BRENT.l2 -0.312650   0.199372  -1.568   0.1175
## CRUDE_WTI.l2   -0.169444   0.127424  -1.330   0.1842
## CRUDE_DUBAI.l2  0.359075   0.168922   2.126   0.0340 *
## CRUDE_BRENT.l3  0.232006   0.205170   1.131   0.2587
## CRUDE_WTI.l3   -0.074396   0.129656  -0.574   0.5664
## CRUDE_DUBAI.l3 -0.162714   0.171766  -0.947   0.3439
## CRUDE_BRENT.l4 -0.030810   0.201810  -0.153   0.8787
## CRUDE_WTI.l4   -0.177593   0.125449  -1.416   0.1575
## CRUDE_DUBAI.l4  0.167068   0.172534   0.968   0.3334
## CRUDE_BRENT.l5 -0.049776   0.194755  -0.256   0.7984
## CRUDE_WTI.l5    0.173164   0.122142   1.418   0.1569
## CRUDE_DUBAI.l5 -0.106108   0.170135  -0.624   0.5331
## CRUDE_BRENT.l6  0.228068   0.190300   1.198   0.2313
## CRUDE_WTI.l6    0.083937   0.119636   0.702   0.4833
## CRUDE_DUBAI.l6 -0.406209   0.168173  -2.415   0.0161 *
## const          0.001447   0.003857   0.375   0.7077
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.08746 on 496 degrees of freedom
## Multiple R-Squared: 0.1365, Adjusted R-squared: 0.1052

```

```

## F-statistic: 4.357 on 18 and 496 DF, p-value: 9.515e-09
##
##
## Estimation results for equation CRUDE_WTI:
## =====
## CRUDE_WTI = CRUDE_BRENT.l1 + CRUDE_WTI.l1 + CRUDE_DUBAI.l1 + CRUDE_BRENT.l2 + CRUDE_WTI
.l2 + CRUDE_DUBAI.l2 + CRUDE_BRENT.l3 + CRUDE_WTI.l3 + CRUDE_DUBAI.l3 + CRUDE_BRENT.l4 + C
RUDE_WTI.l4 + CRUDE_DUBAI.l4 + CRUDE_BRENT.l5 + CRUDE_WTI.l5 + CRUDE_DUBAI.l5 + CRUDE_BREN
T.l6 + CRUDE_WTI.l6 + CRUDE_DUBAI.l6 + const
##
##
## Estimate Std. Error t value Pr(>|t|)
## CRUDE_BRENT.l1 0.174090 0.193697 0.899 0.36921
## CRUDE_WTI.l1 0.001277 0.126640 0.010 0.99196
## CRUDE_DUBAI.l1 0.133123 0.167652 0.794 0.42755
## CRUDE_BRENT.l2 0.104766 0.200056 0.524 0.60073
## CRUDE_WTI.l2 -0.412260 0.127861 -3.224 0.00135 **
## CRUDE_DUBAI.l2 0.202817 0.169502 1.197 0.23205
## CRUDE_BRENT.l3 0.531082 0.205874 2.580 0.01018 *
## CRUDE_WTI.l3 -0.232584 0.130100 -1.788 0.07443 .
## CRUDE_DUBAI.l3 -0.342809 0.172356 -1.989 0.04725 *
## CRUDE_BRENT.l4 0.146350 0.202503 0.723 0.47020
## CRUDE_WTI.l4 -0.321262 0.125879 -2.552 0.01100 *
## CRUDE_DUBAI.l4 0.135275 0.173125 0.781 0.43496
## CRUDE_BRENT.l5 0.136276 0.195423 0.697 0.48592
## CRUDE_WTI.l5 0.035686 0.122561 0.291 0.77104
## CRUDE_DUBAI.l5 -0.184401 0.170719 -1.080 0.28060
## CRUDE_BRENT.l6 0.370974 0.190953 1.943 0.05261 .
## CRUDE_WTI.l6 -0.021890 0.120047 -0.182 0.85539
## CRUDE_DUBAI.l6 -0.425737 0.168749 -2.523 0.01195 *
## const 0.001432 0.003870 0.370 0.71156
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.08776 on 496 degrees of freedom
## Multiple R-Squared: 0.1441, Adjusted R-squared: 0.113
## F-statistic: 4.638 on 18 and 496 DF, p-value: 1.64e-09
##
##
## Estimation results for equation CRUDE_DUBAI:
## =====
## CRUDE_DUBAI = CRUDE_BRENT.l1 + CRUDE_WTI.l1 + CRUDE_DUBAI.l1 + CRUDE_BRENT.l2 + CRUDE_W
TI.l2 + CRUDE_DUBAI.l2 + CRUDE_BRENT.l3 + CRUDE_WTI.l3 + CRUDE_DUBAI.l3 + CRUDE_BRENT.l4 +
CRUDE_WTI.l4 + CRUDE_DUBAI.l4 + CRUDE_BRENT.l5 + CRUDE_WTI.l5 + CRUDE_DUBAI.l5 + CRUDE_BRE
NT.l6 + CRUDE_WTI.l6 + CRUDE_DUBAI.l6 + const
##
##
## Estimate Std. Error t value Pr(>|t|)
## CRUDE_BRENT.l1 -0.078028 0.193563 -0.403 0.68704
## CRUDE_WTI.l1 0.174951 0.126552 1.382 0.16746
## CRUDE_DUBAI.l1 0.243103 0.167536 1.451 0.14740
## CRUDE_BRENT.l2 -0.150141 0.199917 -0.751 0.45300
## CRUDE_WTI.l2 -0.119065 0.127772 -0.932 0.35187
## CRUDE_DUBAI.l2 0.148301 0.169384 0.876 0.38171
## CRUDE_BRENT.l3 0.273722 0.205731 1.330 0.18397
## CRUDE_WTI.l3 -0.007291 0.130010 -0.056 0.95530

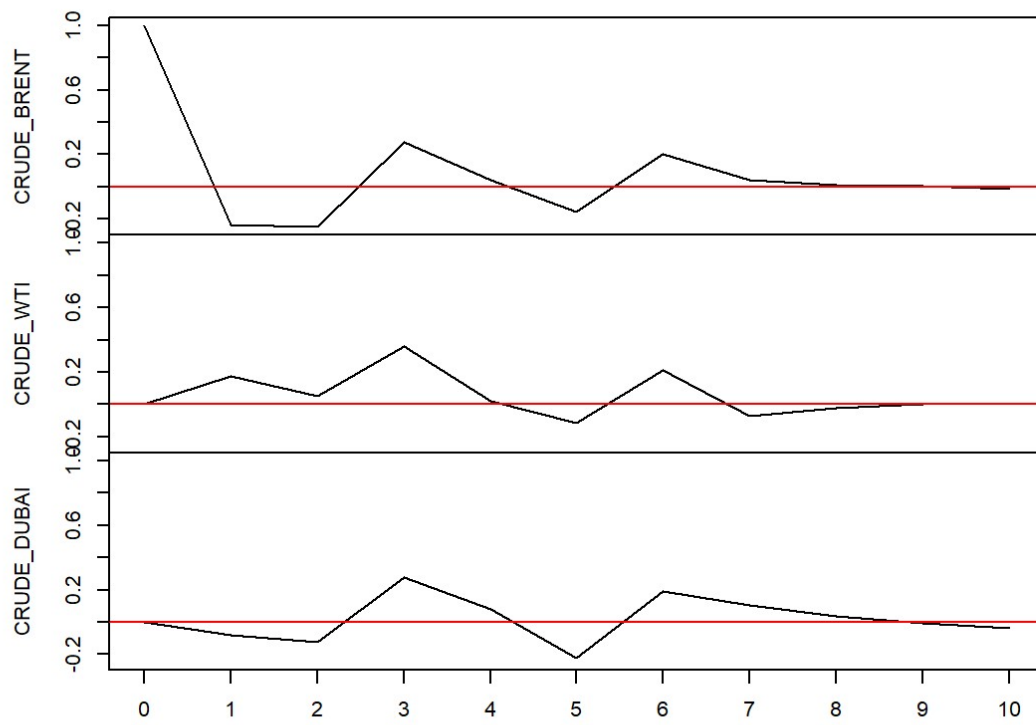
```

```

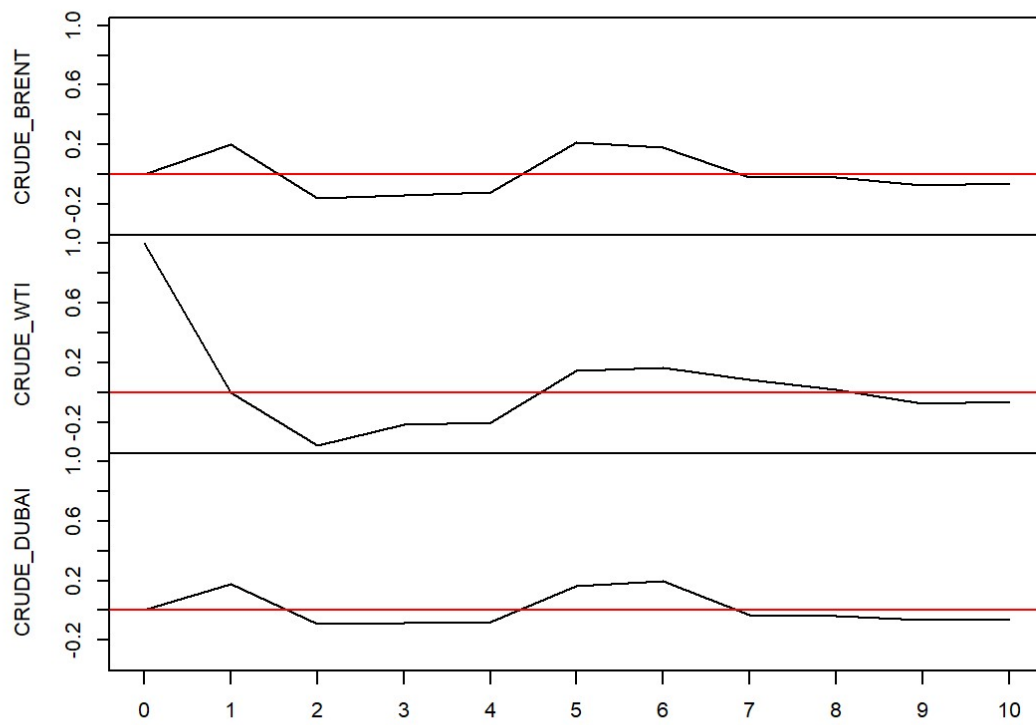
## CRUDE_DUBAI.l3 -0.287148  0.172236 -1.667  0.09611 .
## CRUDE_BRENT.l4  0.002331  0.202362  0.012  0.99082
## CRUDE_WTI.l4   -0.088454  0.125792 -0.703  0.48227
## CRUDE_DUBAI.l4  0.043049  0.173005  0.249  0.80360
## CRUDE_BRENT.l5 -0.144136  0.195287 -0.738  0.46082
## CRUDE_WTI.l5   0.177209  0.122476  1.447  0.14856
## CRUDE_DUBAI.l5 -0.027749  0.170601 -0.163  0.87086
## CRUDE_BRENT.l6  0.197715  0.190820  1.036  0.30065
## CRUDE_WTI.l6   0.139617  0.119963  1.164  0.24505
## CRUDE_DUBAI.l6 -0.453892  0.168632 -2.692  0.00735 **
## const          0.001581  0.003867  0.409  0.68287
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.08769 on 496 degrees of freedom
## Multiple R-Squared:  0.152,    Adjusted R-squared:  0.1212
## F-statistic: 4.938 on 18 and 496 DF,  p-value: 2.49e-10
##
##
## Covariance matrix of residuals:
##          CRUDE_BRENT CRUDE_WTI CRUDE_DUBAI
## CRUDE_BRENT    0.007648  0.007180    0.007389
## CRUDE_WTI      0.007180  0.007701    0.007042
## CRUDE_DUBAI    0.007389  0.007042    0.007690
##
## Correlation matrix of residuals:
##          CRUDE_BRENT CRUDE_WTI CRUDE_DUBAI
## CRUDE_BRENT    1.0000    0.9356    0.9635
## CRUDE_WTI      0.9356    1.0000    0.9150
## CRUDE_DUBAI    0.9635    0.9150    1.0000

```

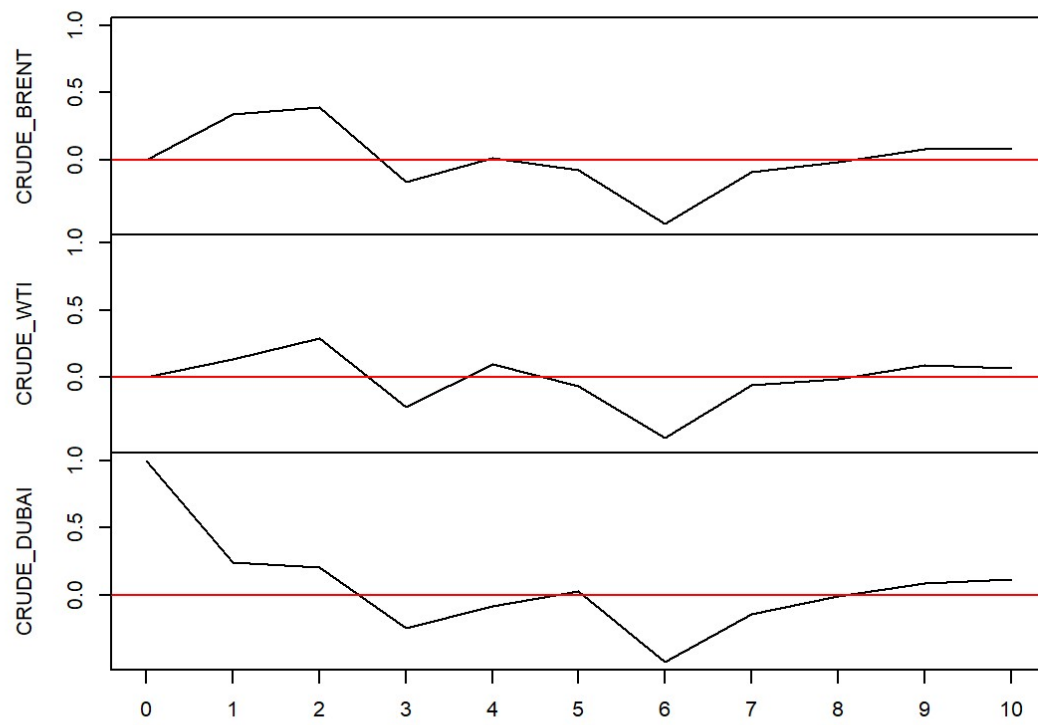

Impulse Response from CRUDE_BRENT



Impulse Response from CRUDE_WTI



Impulse Response from CRUDE_DUBAI



```

##
## Conducting Johansen Cointegration Test:
## Johansen cointegration test summary:
##
## #####
## # Johansen-Procedure #
## #####
##
## Test type: trace statistic , without linear trend and constant in cointegration
##
## Eigenvalues (lambda):
## [1] 4.870423e-02 4.458296e-02 3.690897e-03 1.387779e-17
##
## Values of teststatistic and critical values of test:
##
##          test 10pct  5pct  1pct
## r <= 2 |   1.90   7.52   9.24 12.97
## r <= 1 |  25.39  17.85  19.96 24.60
## r = 0  |  51.11  32.00  34.91 41.07
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##          CRUDE_BRENT.17 CRUDE_WTI.17 CRUDE_DUBAI.17    constant
## CRUDE_BRENT.17          1.00000    1.00000000    1.000000    1.000000
## CRUDE_WTI.17          -53.42506   -0.45837008   -5.755309   20.5568801
## CRUDE_DUBAI.17         45.01413   -0.55926679    6.928170  -27.4336773
## constant              29.89826    0.03186237   -8.077532    0.2024375
##
## Weights W:
## (This is the loading matrix)
##
##          CRUDE_BRENT.17 CRUDE_WTI.17 CRUDE_DUBAI.17    constant
## CRUDE_BRENT.d    0.0005831579 -0.208917057   -0.002313875  1.088682e-17
## CRUDE_WTI.d      0.0017925032  0.001761348   -0.002333404  2.138079e-18
## CRUDE_DUBAI.d   -0.0005933686 -0.079630319   -0.002414131  4.998453e-18
##
## Estimated cointegration rank (5% significance): 2
##
## Fitting VECM Model:
## VECM fit summary:
## #####
## ###Model VECM
## #####
## Full sample size: 522    End sample size: 515
## Number of variables: 3    Number of estimated slope parameters 63
## AIC -9940.038    BIC -9664.167    SSR 11.39081
## Cointegrating vector (estimated by ML):
##    CRUDE_BRENT    CRUDE_WTI CRUDE_DUBAI
## r1          1 5.551115e-17  -0.9536389
## r2          0 1.000000e+00  -0.8603891
##
##
##
##          ECT1          ECT2

```

```

## Equation CRUDE_BRENT -0.2086(0.1502)      0.0644(0.0950)
## Equation CRUDE_WTI    0.0032(0.1507)      -0.0968(0.0954)
## Equation CRUDE_DUBAI -0.0805(0.1508)      0.0679(0.0955)
##                               Intercept      CRUDE_BRENT -1      CRUDE_WTI -1
## Equation CRUDE_BRENT 0.0125(0.0372)      -0.0882(0.2222)      0.1564(0.1424)
## Equation CRUDE_WTI    0.0554(0.0374)      0.1554(0.2230)      0.0687(0.1429)
## Equation CRUDE_DUBAI -0.0185(0.0374)      -0.0140(0.2232)      0.1278(0.1430)
##                               CRUDE_DUBAI -1      CRUDE_BRENT -2      CRUDE_WTI -2
## Equation CRUDE_BRENT 0.2326(0.1809)      -0.1798(0.2217)      -0.2051(0.1397)
## Equation CRUDE_WTI    0.0960(0.1815)      0.0969(0.2225)      -0.3515(0.1402)*
## Equation CRUDE_DUBAI 0.2211(0.1817)      -0.0962(0.2227)      -0.1599(0.1403)
##                               CRUDE_DUBAI -2      CRUDE_BRENT -3      CRUDE_WTI -3
## Equation CRUDE_BRENT 0.2607(0.1818)      0.3428(0.2203)      -0.1053(0.1379)
## Equation CRUDE_WTI    0.1608(0.1824)      0.5267(0.2211)*      -0.1839(0.1384)
## Equation CRUDE_DUBAI 0.1299(0.1826)      0.3177(0.2213)      -0.0409(0.1385)
##                               CRUDE_DUBAI -3      CRUDE_BRENT -4      CRUDE_WTI -4
## Equation CRUDE_BRENT -0.2448(0.1806)      0.0620(0.2122)      -0.2006(0.1307)
## Equation CRUDE_WTI    -0.3775(0.1813)*      0.1491(0.2130)      -0.2822(0.1312)*
## Equation CRUDE_DUBAI -0.3027(0.1814).      0.0362(0.2132)      -0.1147(0.1313)
##                               CRUDE_DUBAI -4      CRUDE_BRENT -5      CRUDE_WTI -5
## Equation CRUDE_BRENT 0.0973(0.1790)      0.0210(0.2010)      0.1562(0.1251)
## Equation CRUDE_WTI    0.1027(0.1796)      0.1399(0.2017)      0.0644(0.1255)
## Equation CRUDE_DUBAI 0.0312(0.1798)      -0.1189(0.2019)      0.1578(0.1257)
##                               CRUDE_DUBAI -5      CRUDE_BRENT -6      CRUDE_WTI -6
## Equation CRUDE_BRENT -0.1613(0.1743)      0.2858(0.1944)      0.0770(0.1215)
## Equation CRUDE_WTI    -0.2098(0.1749)      0.3789(0.1951).      0.0046(0.1219)
## Equation CRUDE_DUBAI -0.0372(0.1750)      0.2161(0.1953)      0.1251(0.1220)
##                               CRUDE_DUBAI -6
## Equation CRUDE_BRENT -0.4576(0.1718)**
## Equation CRUDE_WTI    -0.4498(0.1724)**
## Equation CRUDE_DUBAI -0.4625(0.1726)**

```

```
do_analysis(data_raw, "Natural Gas", natural_gas_cols)
```

```

##
## =====
## Analyzing Natural Gas Commodities
## =====
## Rows before cleaning: 786. After omitting NA: 582. Removed: 204
##
## ADF Test for Stationarity:
## ADF Test Results (on differenced log prices):

```

```
## NGAS_US      : ADF Stat = -10.139, p-value = 0.0100 --> Stationary
```

```
## NGAS_EUR     : ADF Stat = -6.978, p-value = 0.0100 --> Stationary
```

```
## NGAS_JP      : ADF Stat = -7.501, p-value = 0.0100 --> Stationary
```

```

## Fitting VAR Model:
## Lag order selection (AIC, BIC, HQIC):
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      12      4      1      12
## Selected lag order by AIC: 12
## VAR Model summary:
## [1] "VAR Model object created. Checking its structure:"
##
## VAR Estimation Results:
## =====
## Endogenous variables: NGAS_US, NGAS_EUR, NGAS_JP
## Deterministic variables: const
## Sample size: 569
## Log Likelihood: 2036.012
## Roots of the characteristic polynomial:
## 0.9221 0.9221 0.9021 0.9021 0.8877 0.8877 0.8858 0.8858 0.8828 0.8828 0.8807 0.8807 0.8
71 0.871 0.8633 0.8633 0.859 0.859 0.8574 0.8574 0.8571 0.8571 0.8403 0.8403 0.8386 0.8332
0.8332 0.829 0.829 0.8058 0.8058 0.7919 0.7919 0.7373 0.7373 0.5045
## Call:
## VAR(y = diff_df, p = selected_lag, type = "const")
##
##
## Estimation results for equation NGAS_US:
## =====
## NGAS_US = NGAS_US.l1 + NGAS_EUR.l1 + NGAS_JP.l1 + NGAS_US.l2 + NGAS_EUR.l2 + NGAS_JP.l2
+ NGAS_US.l3 + NGAS_EUR.l3 + NGAS_JP.l3 + NGAS_US.l4 + NGAS_EUR.l4 + NGAS_JP.l4 + NGAS_US.
15 + NGAS_EUR.l5 + NGAS_JP.l5 + NGAS_US.l6 + NGAS_EUR.l6 + NGAS_JP.l6 + NGAS_US.l7 + NGAS_
EUR.l7 + NGAS_JP.l7 + NGAS_US.l8 + NGAS_EUR.l8 + NGAS_JP.l8 + NGAS_US.l9 + NGAS_EUR.l9 + N
GAS_JP.l9 + NGAS_US.l10 + NGAS_EUR.l10 + NGAS_JP.l10 + NGAS_US.l11 + NGAS_EUR.l11 + NGAS_J
P.l11 + NGAS_US.l12 + NGAS_EUR.l12 + NGAS_JP.l12 + const
##
##              Estimate Std. Error t value Pr(>|t|)
## NGAS_US.l1    0.014786    0.043461   0.340  0.73383
## NGAS_EUR.l1    0.285969    0.068097   4.199 3.14e-05 ***
## NGAS_JP.l1   -0.060570    0.125139  -0.484  0.62857
## NGAS_US.l2   -0.100441    0.043514  -2.308  0.02137 *
## NGAS_EUR.l2    0.220377    0.069602   3.166  0.00163 **
## NGAS_JP.l2   -0.084962    0.126647  -0.671  0.50260
## NGAS_US.l3   -0.099665    0.044159  -2.257  0.02442 *
## NGAS_EUR.l3   -0.066513    0.070078  -0.949  0.34298
## NGAS_JP.l3    0.215998    0.126575   1.706  0.08850 .
## NGAS_US.l4   -0.034000    0.043946  -0.774  0.43946
## NGAS_EUR.l4    0.086408    0.070520   1.225  0.22101
## NGAS_JP.l4   -0.192942    0.126866  -1.521  0.12890
## NGAS_US.l5   -0.113014    0.044150  -2.560  0.01075 *
## NGAS_EUR.l5    0.014942    0.071478   0.209  0.83450
## NGAS_JP.l5    0.078706    0.129806   0.606  0.54455
## NGAS_US.l6   -0.003808    0.044452  -0.086  0.93176
## NGAS_EUR.l6   -0.046754    0.071372  -0.655  0.51270
## NGAS_JP.l6   -0.116008    0.129952  -0.893  0.37243
## NGAS_US.l7   -0.116217    0.044811  -2.594  0.00976 **
## NGAS_EUR.l7   -0.039406    0.071191  -0.554  0.58013
## NGAS_JP.l7    0.044006    0.130067   0.338  0.73524
## NGAS_US.l8   -0.064162    0.044805  -1.432  0.15272

```

```

## NGAS_EUR.l18    0.196523    0.071184    2.761    0.00597 **
## NGAS_JP.l18     -0.046749    0.130304   -0.359    0.71991
## NGAS_US.l19     -0.142119    0.044932   -3.163    0.00165 **
## NGAS_EUR.l19     0.035568    0.071079    0.500    0.61700
## NGAS_JP.l19      0.161664    0.126619    1.277    0.20224
## NGAS_US.l110     0.015010    0.045144    0.332    0.73965
## NGAS_EUR.l110    -0.005087    0.070320   -0.072    0.94236
## NGAS_JP.l110     -0.020364    0.126669   -0.161    0.87234
## NGAS_US.l111     0.099457    0.044944    2.213    0.02733 *
## NGAS_EUR.l111     0.042014    0.070703    0.594    0.55261
## NGAS_JP.l111     -0.187640    0.123744   -1.516    0.13002
## NGAS_US.l112     -0.117248    0.045103   -2.600    0.00959 **
## NGAS_EUR.l112    -0.105035    0.070916   -1.481    0.13917
## NGAS_JP.l112      0.022788    0.119836    0.190    0.84926
## const            0.002150    0.005312    0.405    0.68588
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.1255 on 532 degrees of freedom
## Multiple R-Squared: 0.1385, Adjusted R-squared: 0.0802
## F-statistic: 2.376 on 36 and 532 DF, p-value: 2.031e-05
##
##
## Estimation results for equation NGAS_EUR:
## =====
## NGAS_EUR = NGAS_US.l1 + NGAS_EUR.l1 + NGAS_JP.l1 + NGAS_US.l2 + NGAS_EUR.l2 + NGAS_JP.l
2 + NGAS_US.l3 + NGAS_EUR.l3 + NGAS_JP.l3 + NGAS_US.l4 + NGAS_EUR.l4 + NGAS_JP.l4 + NGAS_U
S.l5 + NGAS_EUR.l5 + NGAS_JP.l5 + NGAS_US.l6 + NGAS_EUR.l6 + NGAS_JP.l6 + NGAS_US.l7 + NGA
S_EUR.l7 + NGAS_JP.l7 + NGAS_US.l8 + NGAS_EUR.l8 + NGAS_JP.l8 + NGAS_US.l9 + NGAS_EUR.l9 +
NGAS_JP.l9 + NGAS_US.l10 + NGAS_EUR.l10 + NGAS_JP.l10 + NGAS_US.l11 + NGAS_EUR.l11 + NGAS_
JP.l11 + NGAS_US.l12 + NGAS_EUR.l12 + NGAS_JP.l12 + const
##
##
##              Estimate Std. Error t value Pr(>|t|)
## NGAS_US.l1    0.086903    0.028376    3.063 0.002306 **
## NGAS_EUR.l1    0.204639    0.044462    4.603 5.22e-06 ***
## NGAS_JP.l1    -0.098643    0.081705   -1.207 0.227853
## NGAS_US.l2     0.063782    0.028411    2.245 0.025182 *
## NGAS_EUR.l2     0.030170    0.045444    0.664 0.507047
## NGAS_JP.l2    -0.141068    0.082690   -1.706 0.088595 .
## NGAS_US.l3     0.028945    0.028832    1.004 0.315877
## NGAS_EUR.l3     0.117359    0.045755    2.565 0.010592 *
## NGAS_JP.l3     0.209555    0.082643    2.536 0.011508 *
## NGAS_US.l4     0.062686    0.028693    2.185 0.029345 *
## NGAS_EUR.l4    -0.022680    0.046044   -0.493 0.622523
## NGAS_JP.l4     0.104906    0.082833    1.266 0.205899
## NGAS_US.l5     0.041760    0.028826    1.449 0.148023
## NGAS_EUR.l5    -0.007471    0.046669   -0.160 0.872869
## NGAS_JP.l5     -0.074867    0.084752   -0.883 0.377442
## NGAS_US.l6     0.001590    0.029023    0.055 0.956334
## NGAS_EUR.l6    -0.109571    0.046600   -2.351 0.019071 *
## NGAS_JP.l6     -0.159069    0.084848   -1.875 0.061374 .
## NGAS_US.l7     -0.010908    0.029258   -0.373 0.709416
## NGAS_EUR.l7    -0.078979    0.046482   -1.699 0.089876 .
## NGAS_JP.l7     -0.020587    0.084923   -0.242 0.808550

```

```

## NGAS_US.l8      0.048256    0.029254    1.650 0.099627 .
## NGAS_EUR.l8     0.016700    0.046477    0.359 0.719507
## NGAS_JP.l8      0.097876    0.085078    1.150 0.250482
## NGAS_US.l9     -0.046161    0.029337   -1.573 0.116202
## NGAS_EUR.l9     0.086989    0.046409    1.874 0.061422 .
## NGAS_JP.l9     -0.093539    0.082672   -1.131 0.258373
## NGAS_US.l10     0.074863    0.029475    2.540 0.011374 *
## NGAS_EUR.l10    0.022519    0.045913    0.490 0.624009
## NGAS_JP.l10     0.051355    0.082705    0.621 0.534905
## NGAS_US.l11     0.006681    0.029345    0.228 0.819979
## NGAS_EUR.l11    0.170650    0.046163    3.697 0.000241 ***
## NGAS_JP.l11    -0.236357    0.080794   -2.925 0.003587 **
## NGAS_US.l12    -0.057966    0.029448   -1.968 0.049540 *
## NGAS_EUR.l12    0.059697    0.046302    1.289 0.197860
## NGAS_JP.l12     0.029560    0.078243    0.378 0.705734
## const           0.001577    0.003468    0.455 0.649559
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.08192 on 532 degrees of freedom
## Multiple R-Squared:  0.2325, Adjusted R-squared:  0.1806
## F-statistic: 4.476 on 36 and 532 DF, p-value: 4.767e-15
##
##
## Estimation results for equation NGAS_JP:
## =====
## NGAS_JP = NGAS_US.l1 + NGAS_EUR.l1 + NGAS_JP.l1 + NGAS_US.l2 + NGAS_EUR.l2 + NGAS_JP.l2
+ NGAS_US.l3 + NGAS_EUR.l3 + NGAS_JP.l3 + NGAS_US.l4 + NGAS_EUR.l4 + NGAS_JP.l4 + NGAS_US.
l5 + NGAS_EUR.l5 + NGAS_JP.l5 + NGAS_US.l6 + NGAS_EUR.l6 + NGAS_JP.l6 + NGAS_US.l7 + NGAS_
EUR.l7 + NGAS_JP.l7 + NGAS_US.l8 + NGAS_EUR.l8 + NGAS_JP.l8 + NGAS_US.l9 + NGAS_EUR.l9 + N
GAS_JP.l9 + NGAS_US.l10 + NGAS_EUR.l10 + NGAS_JP.l10 + NGAS_US.l11 + NGAS_EUR.l11 + NGAS_J
P.l11 + NGAS_US.l12 + NGAS_EUR.l12 + NGAS_JP.l12 + const
##
##              Estimate Std. Error t value Pr(>|t|)
## NGAS_US.l1    0.0135052   0.0154001    0.877  0.38091
## NGAS_EUR.l1    0.1048927   0.0241297    4.347 1.65e-05 ***
## NGAS_JP.l1     0.1409947   0.0443422    3.180  0.00156 **
## NGAS_US.l2     0.0391733   0.0154190    2.541  0.01135 *
## NGAS_EUR.l2     0.0651034   0.0246629    2.640  0.00854 **
## NGAS_JP.l2    -0.0055550   0.0448768   -0.124  0.90153
## NGAS_US.l3     0.0030453   0.0156476    0.195  0.84577
## NGAS_EUR.l3    -0.0214128   0.0248316   -0.862  0.38890
## NGAS_JP.l3     0.0417499   0.0448510    0.931  0.35235
## NGAS_US.l4     0.0372020   0.0155718    2.389  0.01724 *
## NGAS_EUR.l4     0.1045072   0.0249885    4.182 3.38e-05 ***
## NGAS_JP.l4    -0.1793728   0.0449544   -3.990 7.53e-05 ***
## NGAS_US.l5    -0.0007412   0.0156444   -0.047  0.96223
## NGAS_EUR.l5     0.0223911   0.0253279    0.884  0.37707
## NGAS_JP.l5    -0.0011025   0.0459959   -0.024  0.98089
## NGAS_US.l6     0.0215452   0.0157512    1.368  0.17194
## NGAS_EUR.l6     0.0202418   0.0252902    0.800  0.42385
## NGAS_JP.l6    -0.0294384   0.0460479   -0.639  0.52290
## NGAS_US.l7     0.0164585   0.0158785    1.037  0.30043
## NGAS_EUR.l7    -0.0179184   0.0252260   -0.710  0.47782

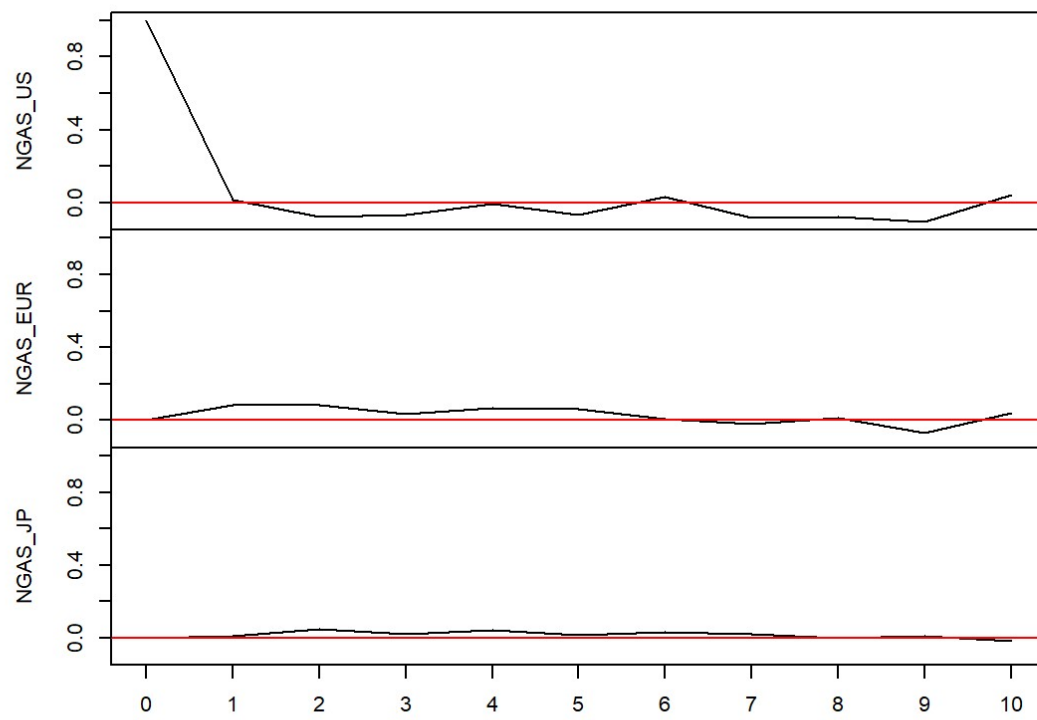
```

```

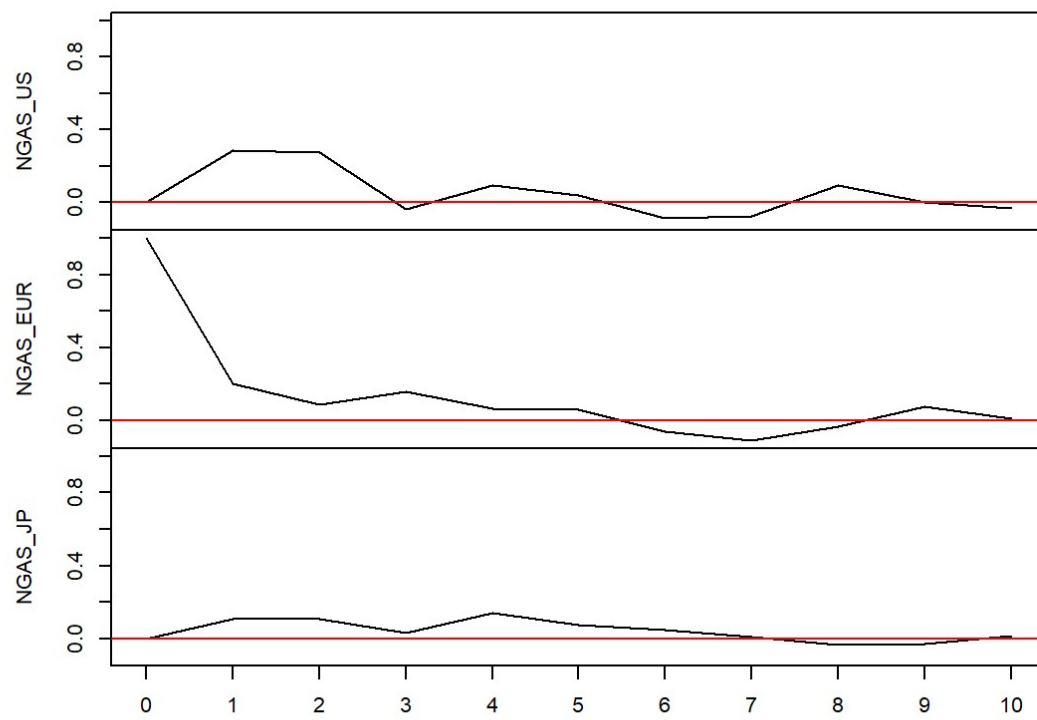
## NGAS_JP.17      0.0383090  0.0460885   0.831  0.40623
## NGAS_US.18      0.0034597  0.0158764   0.218  0.82758
## NGAS_EUR.18     -0.0191763  0.0252236  -0.760  0.44744
## NGAS_JP.18      -0.0536230  0.0461724  -1.161  0.24601
## NGAS_US.19      0.0170363  0.0159214   1.070  0.28510
## NGAS_EUR.19     -0.0090634  0.0251864  -0.360  0.71910
## NGAS_JP.19      -0.0216154  0.0448667  -0.482  0.63017
## NGAS_US.110     0.0002577  0.0159966   0.016  0.98715
## NGAS_EUR.110    0.0326057  0.0249176   1.309  0.19125
## NGAS_JP.110     0.0364132  0.0448845   0.811  0.41758
## NGAS_US.111     -0.0065897  0.0159256  -0.414  0.67920
## NGAS_EUR.111    0.0113203  0.0250532   0.452  0.65156
## NGAS_JP.111     -0.0589558  0.0438479  -1.345  0.17934
## NGAS_US.112     0.0135395  0.0159818   0.847  0.39727
## NGAS_EUR.112    0.0042620  0.0251288   0.170  0.86538
## NGAS_JP.112     0.1364965  0.0424633   3.214  0.00139 **
## const           0.0009432  0.0018822   0.501  0.61651
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.04446 on 532 degrees of freedom
## Multiple R-Squared: 0.2702, Adjusted R-squared: 0.2208
## F-statistic: 5.472 on 36 and 532 DF, p-value: < 2.2e-16
##
##
##
## Covariance matrix of residuals:
##           NGAS_US  NGAS_EUR  NGAS_JP
## NGAS_US  0.0157431 0.0011088 0.0007471
## NGAS_EUR 0.0011088 0.0067113 0.0008967
## NGAS_JP  0.0007471 0.0008967 0.0019767
##
## Correlation matrix of residuals:
##           NGAS_US  NGAS_EUR  NGAS_JP
## NGAS_US   1.0000   0.1079   0.1339
## NGAS_EUR   0.1079   1.0000   0.2462
## NGAS_JP    0.1339   0.2462   1.0000

```

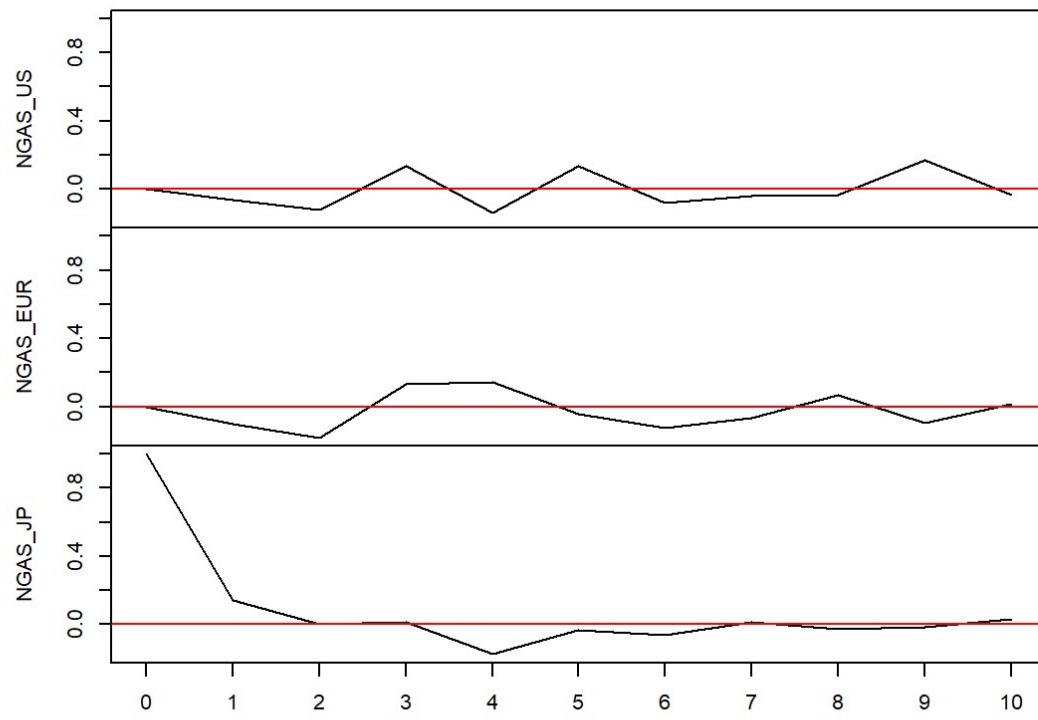

Impulse Response from NGAS_US



Impulse Response from NGAS_EUR



Impulse Response from NGAS_JP



```

##
## Conducting Johansen Cointegration Test:
## Johansen cointegration test summary:
##
## #####
## # Johansen-Procedure #
## #####
##
## Test type: trace statistic , without linear trend and constant in cointegration
##
## Eigenvalues (lambda):
## [1] 5.507160e-02 1.513716e-02 3.483500e-03 2.332056e-19
##
## Values of teststatistic and critical values of test:
##
##          test 10pct  5pct  1pct
## r <= 2 |   1.99   7.52   9.24 12.97
## r <= 1 |  10.66  17.85  19.96 24.60
## r = 0  |  42.90  32.00  34.91 41.07
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##          NGAS_US.l13 NGAS_EUR.l13 NGAS_JP.l13  constant
## NGAS_US.l13    1.000000    1.0000000    1.00000000    1.00000000
## NGAS_EUR.l13   -7.810957   -0.5132700  -0.04071622  -0.04943913
## NGAS_JP.l13    7.351278    0.5252329  -1.43649853  -2.63855598
## constant       -2.136650   -1.2086039   1.98434061   1.27382429
##
## Weights W:
## (This is the loading matrix)
##
##          NGAS_US.l13 NGAS_EUR.l13  NGAS_JP.l13  constant
## NGAS_US.d    0.002690351 -0.031935440  0.0006466772 -8.295841e-18
## NGAS_EUR.d    0.009074567  0.001044396  0.0040372279 -6.884632e-19
## NGAS_JP.d   -0.002664012 -0.001583608  0.0030893365  4.769852e-20
##
## Estimated cointegration rank (5% significance): 1
##
## Fitting VECM Model:
## VECM fit summary:
## #####
## ###Model VECM
## #####
## Full sample size: 582    End sample size: 569
## Number of variables: 3    Number of estimated slope parameters 114
## AIC -8716.491    BIC -8212.601    SSR 12.8616
## Cointegrating vector (estimated by ML):
##    NGAS_US  NGAS_EUR  NGAS_JP
## r1         1 -7.793981 7.337624
##
##
##          ECT          Intercept          NGAS_US -1
## Equation NGAS_US  0.0027(0.0033)  -0.0036(0.0090)  0.0107(0.0438)

```

```

## Equation NGAS_EUR 0.0091(0.0022)*** -0.0182(0.0058)** 0.0730(0.0281)**
## Equation NGAS_JP -0.0027(0.0012)* 0.0068(0.0032)* 0.0176(0.0154)
##          NGAS_EUR -1      NGAS_JP -1      NGAS_US -2
## Equation NGAS_US 0.2962(0.0693)*** -0.0640(0.1253) -0.1043(0.0438)*
## Equation NGAS_EUR 0.2395(0.0446)*** -0.1104(0.0805) 0.0505(0.0281).
## Equation NGAS_JP 0.0946(0.0245)*** 0.1445(0.0442)** 0.0431(0.0155)**
##          NGAS_EUR -2      NGAS_JP -2      NGAS_US -3
## Equation NGAS_US 0.2327(0.0713)** -0.0918(0.1270) -0.1038(0.0445)*
## Equation NGAS_EUR 0.0724(0.0459) -0.1643(0.0816)* 0.0147(0.0286)
## Equation NGAS_JP 0.0526(0.0252)* 0.0013(0.0448) 0.0073(0.0157)
##          NGAS_EUR -3      NGAS_JP -3      NGAS_US -4
## Equation NGAS_US -0.0547(0.0717) 0.2095(0.1269). -0.0377(0.0442)
## Equation NGAS_EUR 0.1578(0.0461)*** 0.1873(0.0815)* 0.0500(0.0284).
## Equation NGAS_JP -0.0334(0.0253) 0.0484(0.0448) 0.0410(0.0156)**
##          NGAS_EUR -4      NGAS_JP -4      NGAS_US -5
## Equation NGAS_US 0.1012(0.0730) -0.1984(0.1271) -0.1161(0.0443)**
## Equation NGAS_EUR 0.0278(0.0469) 0.0862(0.0817) 0.0311(0.0285)
## Equation NGAS_JP 0.0895(0.0257)*** -0.1738(0.0448)*** 0.0024(0.0156)
##          NGAS_EUR -5      NGAS_JP -5      NGAS_US -6
## Equation NGAS_US 0.0277(0.0733) 0.0761(0.1299) -0.0069(0.0446)
## Equation NGAS_EUR 0.0360(0.0471) -0.0838(0.0835) -0.0090(0.0287)
## Equation NGAS_JP 0.0095(0.0259) 0.0016(0.0458) 0.0247(0.0157)
##          NGAS_EUR -6      NGAS_JP -6      NGAS_US -7
## Equation NGAS_US -0.0351(0.0729) -0.1196(0.1301) -0.1202(0.0451)**
## Equation NGAS_EUR -0.0696(0.0469) -0.1713(0.0836)* -0.0246(0.0290)
## Equation NGAS_JP 0.0084(0.0257) -0.0258(0.0459) 0.0205(0.0159)
##          NGAS_EUR -7      NGAS_JP -7      NGAS_US -8
## Equation NGAS_US -0.0308(0.0720) 0.0405(0.1302) -0.0683(0.0451)
## Equation NGAS_EUR -0.0496(0.0463) -0.0326(0.0837) 0.0342(0.0290)
## Equation NGAS_JP -0.0266(0.0254) 0.0419(0.0459) 0.0076(0.0159)
##          NGAS_EUR -8      NGAS_JP -8      NGAS_US -9
## Equation NGAS_US 0.2043(0.0719)** -0.0508(0.1304) -0.1460(0.0452)**
## Equation NGAS_EUR 0.0431(0.0462) 0.0841(0.0838) -0.0593(0.0291)*
## Equation NGAS_JP -0.0270(0.0254) -0.0495(0.0460) 0.0209(0.0160)
##          NGAS_EUR -9      NGAS_JP -9      NGAS_US -10
## Equation NGAS_US 0.0430(0.0717) 0.1624(0.1267) 0.0101(0.0456)
## Equation NGAS_EUR 0.1125(0.0461)* -0.0911(0.0814) 0.0579(0.0293)*
## Equation NGAS_JP -0.0166(0.0253) -0.0223(0.0447) 0.0053(0.0161)
##          NGAS_EUR -10     NGAS_JP -10     NGAS_US -11
## Equation NGAS_US 0.0052(0.0715) -0.0209(0.1267) 0.0961(0.0452)*
## Equation NGAS_EUR 0.0579(0.0460) 0.0494(0.0814) -0.0049(0.0290)
## Equation NGAS_JP 0.0221(0.0252) 0.0370(0.0447) -0.0032(0.0159)
##          NGAS_EUR -11     NGAS_JP -11     NGAS_US -12
## Equation NGAS_US 0.0509(0.0716) -0.1850(0.1238) -0.1196(0.0452)**
## Equation NGAS_EUR 0.2011(0.0460)*** -0.2273(0.0796)** -0.0659(0.0291)*
## Equation NGAS_JP 0.0023(0.0253) -0.0617(0.0437) 0.0159(0.0160)
##          NGAS_EUR -12     NGAS_JP -12
## Equation NGAS_US -0.0936(0.0724) 0.0284(0.1201)
## Equation NGAS_EUR 0.0989(0.0465)* 0.0487(0.0772)
## Equation NGAS_JP -0.0074(0.0255) 0.1308(0.0424)**

```

```
do_analysis(data_raw, "Coal", coal_cols)
```

```
##
## =====
## Analyzing Coal Commodities
## =====
## Rows before cleaning: 786. After omitting NA: 498. Removed: 288
##
## ADF Test for Stationarity:
## ADF Test Results (on differenced log prices):
```

```
## COAL_AUS      : ADF Stat = -7.049, p-value = 0.0100 --> Stationary
```

```
## COAL_SAFRICA  : ADF Stat = -7.025, p-value = 0.0100 --> Stationary
```

```

## Fitting VAR Model:
## Lag order selection (AIC, BIC, HQIC):
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      12      1      1      12
## Selected lag order by AIC: 12
## VAR Model summary:
## [1] "VAR Model object created. Checking its structure:"
##
## VAR Estimation Results:
## =====
## Endogenous variables: COAL_AUS, COAL_SAFRICA
## Deterministic variables: const
## Sample size: 485
## Log Likelihood: 1460.668
## Roots of the characteristic polynomial:
## 0.8872 0.8872 0.8864 0.8864 0.8817 0.8817 0.8594 0.8594 0.8587 0.8587 0.8496 0.8496 0.8
465 0.8465 0.8363 0.8363 0.8305 0.8202 0.8202 0.7972 0.7972 0.7683 0.5349 0.5349
## Call:
## VAR(y = diff_df, p = selected_lag, type = "const")
##
##
## Estimation results for equation COAL_AUS:
## =====
## COAL_AUS = COAL_AUS.l1 + COAL_SAFRICA.l1 + COAL_AUS.l2 + COAL_SAFRICA.l2 + COAL_AUS.l3
+ COAL_SAFRICA.l3 + COAL_AUS.l4 + COAL_SAFRICA.l4 + COAL_AUS.l5 + COAL_SAFRICA.l5 + COAL_A
US.l6 + COAL_SAFRICA.l6 + COAL_AUS.l7 + COAL_SAFRICA.l7 + COAL_AUS.l8 + COAL_SAFRICA.l8 +
COAL_AUS.l9 + COAL_SAFRICA.l9 + COAL_AUS.l10 + COAL_SAFRICA.l10 + COAL_AUS.l11 + COAL_SAFR
ICA.l11 + COAL_AUS.l12 + COAL_SAFRICA.l12 + const
##
##
##              Estimate Std. Error t value Pr(>|t|)
## COAL_AUS.l1      0.105349   0.066441   1.586 0.113516
## COAL_SAFRICA.l1  0.215519   0.065163   3.307 0.001016 **
## COAL_AUS.l2     -0.053591   0.066657  -0.804 0.421825
## COAL_SAFRICA.l2  0.070198   0.065929   1.065 0.287546
## COAL_AUS.l3      0.115998   0.066585   1.742 0.082159 .
## COAL_SAFRICA.l3  0.014796   0.065909   0.224 0.822478
## COAL_AUS.l4     -0.161907   0.066862  -2.422 0.015842 *
## COAL_SAFRICA.l4  0.110768   0.065586   1.689 0.091918 .
## COAL_AUS.l5     -0.054543   0.067268  -0.811 0.417882
## COAL_SAFRICA.l5  0.221015   0.065810   3.358 0.000849 ***
## COAL_AUS.l6     -0.181306   0.067103  -2.702 0.007149 **
## COAL_SAFRICA.l6  0.071922   0.066767   1.077 0.281950
## COAL_AUS.l7      0.009782   0.067132   0.146 0.884210
## COAL_SAFRICA.l7  0.123964   0.066749   1.857 0.063927 .
## COAL_AUS.l8     -0.204803   0.066160  -3.096 0.002084 **
## COAL_SAFRICA.l8  0.107800   0.066171   1.629 0.103975
## COAL_AUS.l9      0.004268   0.066530   0.064 0.948879
## COAL_SAFRICA.l9  0.166563   0.065494   2.543 0.011311 *
## COAL_AUS.l10    -0.122003   0.066988  -1.821 0.069218 .
## COAL_SAFRICA.l10 0.036359   0.065870   0.552 0.581231
## COAL_AUS.l11     0.090657   0.066745   1.358 0.175048
## COAL_SAFRICA.l11 -0.128278   0.065668  -1.953 0.051375 .
## COAL_AUS.l12    -0.046549   0.065560  -0.710 0.478049
## COAL_SAFRICA.l12 0.075571   0.065771   1.149 0.251154

```

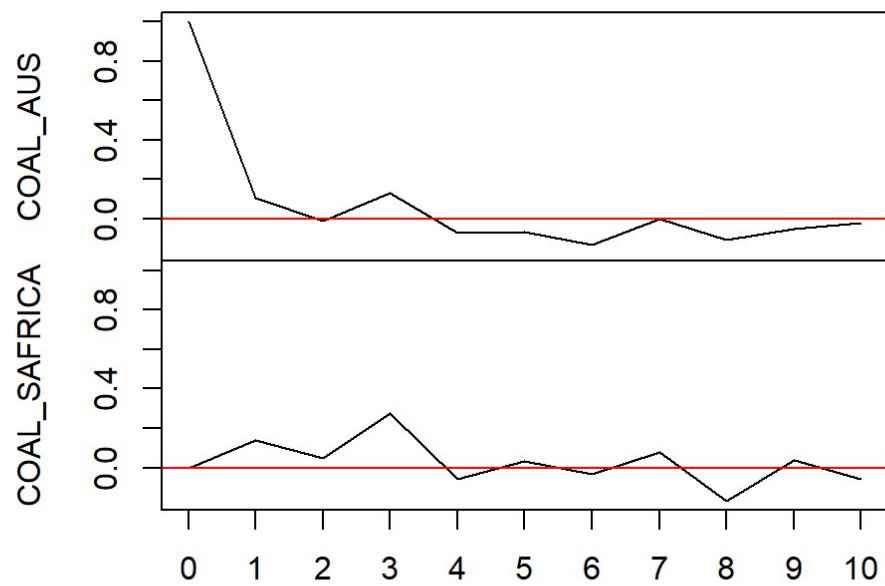
```

## const          0.001518    0.002989    0.508 0.611931
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.06552 on 460 degrees of freedom
## Multiple R-Squared: 0.1939, Adjusted R-squared: 0.1519
## F-statistic: 4.611 on 24 and 460 DF, p-value: 1.674e-11
##
##
## Estimation results for equation COAL_SAFRICA:
## =====
## COAL_SAFRICA = COAL_AUS.l1 + COAL_SAFRICA.l1 + COAL_AUS.l2 + COAL_SAFRICA.l2 + COAL_AUS
.l3 + COAL_SAFRICA.l3 + COAL_AUS.l4 + COAL_SAFRICA.l4 + COAL_AUS.l5 + COAL_SAFRICA.l5 + CO
AL_AUS.l6 + COAL_SAFRICA.l6 + COAL_AUS.l7 + COAL_SAFRICA.l7 + COAL_AUS.l8 + COAL_SAFRICA.l
8 + COAL_AUS.l9 + COAL_SAFRICA.l9 + COAL_AUS.l10 + COAL_SAFRICA.l10 + COAL_AUS.l11 + COAL_
SAFRICA.l11 + COAL_AUS.l12 + COAL_SAFRICA.l12 + const
##
##          Estimate Std. Error t value Pr(>|t|)
## COAL_AUS.l1      0.1410930  0.0675265   2.089  0.03722 *
## COAL_SAFRICA.l1   0.1135185  0.0662276   1.714  0.08719 .
## COAL_AUS.l2       0.0160400  0.0677457   0.237  0.81294
## COAL_SAFRICA.l2   0.0116976  0.0670057   0.175  0.86149
## COAL_AUS.l3       0.2693767  0.0676728   3.981 7.99e-05 ***
## COAL_SAFRICA.l3  -0.0903905  0.0669860  -1.349  0.17787
## COAL_AUS.l4      -0.1249386  0.0679544  -1.839  0.06662 .
## COAL_SAFRICA.l4  -0.0286728  0.0666578  -0.430  0.66729
## COAL_AUS.l5       0.0701002  0.0683673   1.025  0.30574
## COAL_SAFRICA.l5  -0.0027050  0.0668847  -0.040  0.96776
## COAL_AUS.l6      -0.0435678  0.0681989  -0.639  0.52325
## COAL_SAFRICA.l6   0.0052676  0.0678578   0.078  0.93816
## COAL_AUS.l7       0.1419821  0.0682286   2.081  0.03799 *
## COAL_SAFRICA.l7  -0.0588781  0.0678397  -0.868  0.38590
## COAL_AUS.l8      -0.1799528  0.0672407  -2.676  0.00771 **
## COAL_SAFRICA.l8   0.0228229  0.0672525   0.339  0.73449
## COAL_AUS.l9       0.1263572  0.0676169   1.869  0.06230 .
## COAL_SAFRICA.l9  -0.0009993  0.0665639  -0.015  0.98803
## COAL_AUS.l10     -0.0792915  0.0680828  -1.165  0.24477
## COAL_SAFRICA.l10  0.0804651  0.0669457   1.202  0.23000
## COAL_AUS.l11      0.0767892  0.0678357   1.132  0.25823
## COAL_SAFRICA.l11 -0.1108530  0.0667409  -1.661  0.09741 .
## COAL_AUS.l12     -0.1049472  0.0666312  -1.575  0.11593
## COAL_SAFRICA.l12  0.0116045  0.0668453   0.174  0.86225
## const            0.0014692  0.0030383   0.484  0.62892
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.06659 on 460 degrees of freedom
## Multiple R-Squared: 0.1463, Adjusted R-squared: 0.1018
## F-statistic: 3.285 on 24 and 460 DF, p-value: 4.492e-07
##
##
##
## Covariance matrix of residuals:

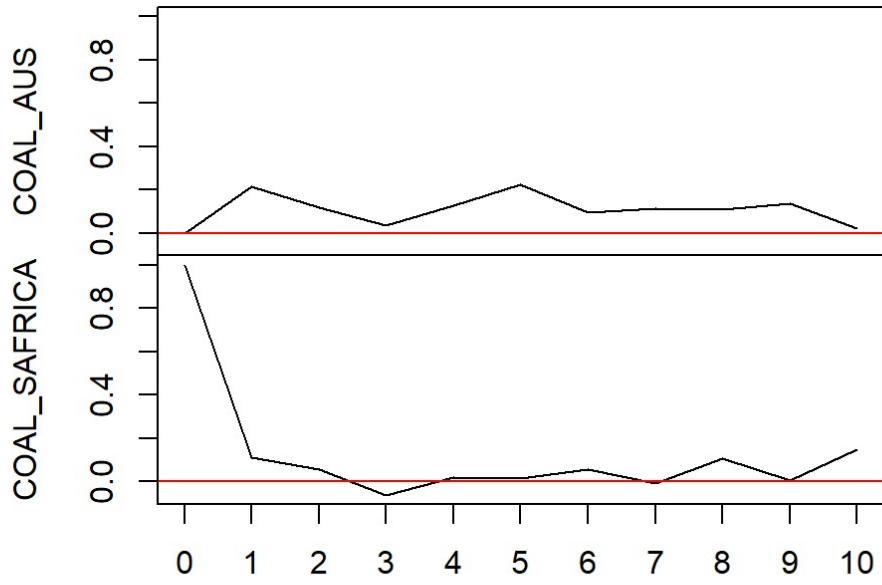
```

```
##          COAL_AUS COAL_SAFRICA
## COAL_AUS    0.004293    0.003132
## COAL_SAFRICA 0.003132    0.004435
##
## Correlation matrix of residuals:
##          COAL_AUS COAL_SAFRICA
## COAL_AUS    1.0000    0.7178
## COAL_SAFRICA 0.7178    1.0000
```

Impulse Response from COAL_AUS



Impulse Response from COAL_SAFRICA



```
##
## Conducting Johansen Cointegration Test:
## Johansen cointegration test summary:
##
## #####
## # Johansen-Procedure #
## #####
##
## Test type: trace statistic , without linear trend and constant in cointegration
##
## Eigenvalues (lambda):
## [1] 1.846252e-02 5.588989e-03 3.469447e-18
##
## Values of teststatistic and critical values of test:
##
##          test 10pct  5pct  1pct
## r <= 1 |   2.72   7.52   9.24 12.97
## r = 0  |  11.76  17.85  19.96 24.60
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##          COAL_AUS.l13 COAL_SAFRICA.l13  constant
## COAL_AUS.l13      1.0000000      1.000000  1.0000000
## COAL_SAFRICA.l13 -0.9988350     -1.817923 -0.4312404
## constant        -0.1011279      3.224413 -1.0300191
##
## Weights W:
## (This is the loading matrix)
##
##          COAL_AUS.l13 COAL_SAFRICA.l13  constant
## COAL_AUS.d      -0.07348542     0.004485602 -4.963155e-17
## COAL_SAFRICA.d  -0.02863929     0.009875704 -2.884946e-17
##
## Estimated cointegration rank (5% significance): 0
##
## No significant cointegration detected; skipping VECM.
```

```
do_analysis(data_raw, "Metals", metals_cols)
```

```
##
## =====
## Analyzing Metals Commodities
## =====
## Rows before cleaning: 786. After omitting NA: 786. Removed: 0
##
## ADF Test for Stationarity:
## ADF Test Results (on differenced log prices):
```

```
## GOLD          : ADF Stat = -6.621, p-value = 0.0100 --> Stationary
```

SILVER : ADF Stat = -8.298, p-value = 0.0100 --> Stationary

COPPER : ADF Stat = -9.003, p-value = 0.0100 --> Stationary

ALUMINUM : ADF Stat = -8.433, p-value = 0.0100 --> Stationary

```

## Fitting VAR Model:
## Lag order selection (AIC, BIC, HQIC):
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      2      1      1      2
## Selected lag order by AIC: 2
## VAR Model summary:
## [1] "VAR Model object created. Checking its structure:"
##
## VAR Estimation Results:
## =====
## Endogenous variables: GOLD, SILVER, COPPER, ALUMINUM
## Deterministic variables: const
## Sample size: 783
## Log Likelihood: 5110.986
## Roots of the characteristic polynomial:
## 0.3907 0.3907 0.35 0.35 0.1876 0.1795 0.1795 0.01082
## Call:
## VAR(y = diff_df, p = selected_lag, type = "const")
##
##
## Estimation results for equation GOLD:
## =====
## GOLD = GOLD.l1 + SILVER.l1 + COPPER.l1 + ALUMINUM.l1 + GOLD.l2 + SILVER.l2 + COPPER.l2
+ ALUMINUM.l2 + const
##
##           Estimate Std. Error t value Pr(>|t|)
## GOLD.l1      0.297454   0.049837   5.969 3.64e-09 ***
## SILVER.l1     0.008923   0.030380   0.294  0.7691
## COPPER.l1    -0.011737   0.027272  -0.430  0.6671
## ALUMINUM.l1  -0.039639   0.036007  -1.101  0.2713
## GOLD.l2     -0.046275   0.049670  -0.932  0.3518
## SILVER.l2    -0.058350   0.030381  -1.921  0.0551 .
## COPPER.l2    -0.021773   0.027351  -0.796  0.4262
## ALUMINUM.l2  0.058535   0.036035   1.624  0.1047
## const        0.004657   0.001516   3.072  0.0022 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.04175 on 774 degrees of freedom
## Multiple R-Squared: 0.09253, Adjusted R-squared: 0.08315
## F-statistic: 9.865 on 8 and 774 DF, p-value: 4.029e-13
##
##
## Estimation results for equation SILVER:
## =====
## SILVER = GOLD.l1 + SILVER.l1 + COPPER.l1 + ALUMINUM.l1 + GOLD.l2 + SILVER.l2 + COPPER.l
2 + ALUMINUM.l2 + const
##
##           Estimate Std. Error t value Pr(>|t|)
## GOLD.l1      0.074837   0.083547   0.896  0.37066
## SILVER.l1     0.242559   0.050930   4.763 2.28e-06 ***
## COPPER.l1     0.036271   0.045720   0.793  0.42782
## ALUMINUM.l1  -0.022422   0.060363  -0.371  0.71040

```

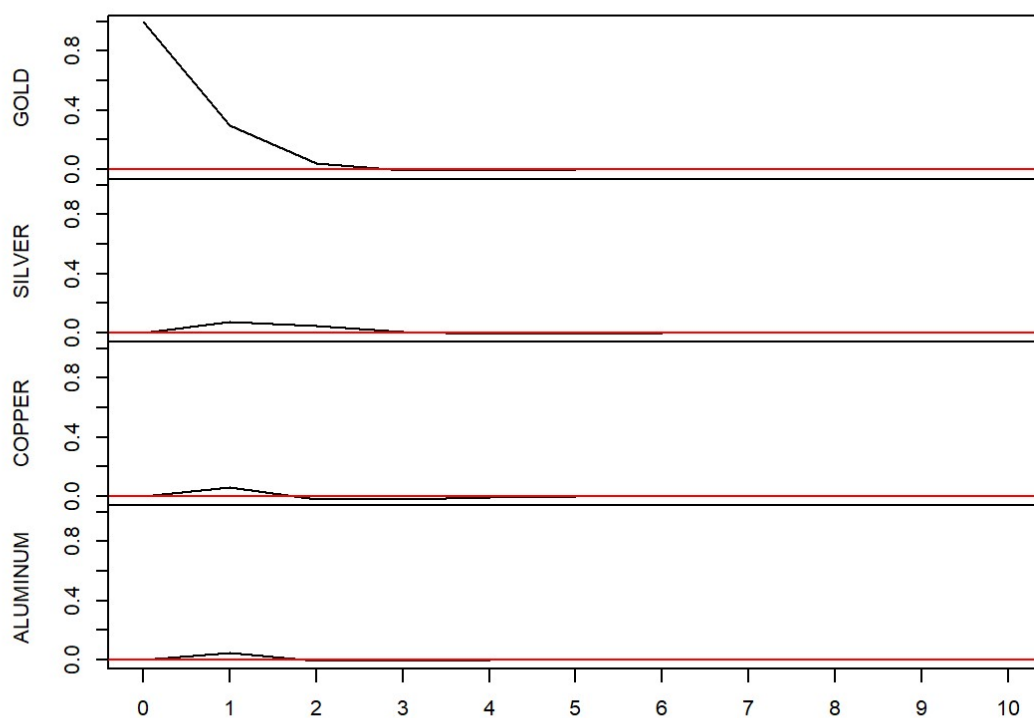
```

## GOLD.l2      0.007524    0.083268    0.090  0.92803
## SILVER.l2    -0.145513    0.050931   -2.857  0.00439 **
## COPPER.l2    -0.023497    0.045852   -0.512  0.60848
## ALUMINUM.l2  0.070921    0.060410    1.174  0.24075
## const       0.003631    0.002542    1.429  0.15349
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.06999 on 774 degrees of freedom
## Multiple R-Squared: 0.08063, Adjusted R-squared: 0.07113
## F-statistic: 8.485 on 8 and 774 DF, p-value: 4.176e-11
##
##
## Estimation results for equation COPPER:
## =====
## COPPER = GOLD.l1 + SILVER.l1 + COPPER.l1 + ALUMINUM.l1 + GOLD.l2 + SILVER.l2 + COPPER.l
2 + ALUMINUM.l2 + const
##
##           Estimate Std. Error t value Pr(>|t|)
## GOLD.l1      0.057892   0.073670   0.786  0.43221
## SILVER.l1     0.042002   0.044909   0.935  0.34994
## COPPER.l1     0.363039   0.040315   9.005 < 2e-16 ***
## ALUMINUM.l1  -0.093270   0.053227  -1.752  0.08011 .
## GOLD.l2      -0.059106   0.073424  -0.805  0.42107
## SILVER.l2     0.053013   0.044910   1.180  0.23819
## COPPER.l2    -0.109037   0.040431  -2.697  0.00715 **
## ALUMINUM.l2  0.017195   0.053268   0.323  0.74693
## const        0.002290   0.002241   1.022  0.30717
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.06172 on 774 degrees of freedom
## Multiple R-Squared: 0.1261, Adjusted R-squared: 0.117
## F-statistic: 13.96 on 8 and 774 DF, p-value: < 2.2e-16
##
##
## Estimation results for equation ALUMINUM:
## =====
## ALUMINUM = GOLD.l1 + SILVER.l1 + COPPER.l1 + ALUMINUM.l1 + GOLD.l2 + SILVER.l2 + COPPER
.l2 + ALUMINUM.l2 + const
##
##           Estimate Std. Error t value Pr(>|t|)
## GOLD.l1      0.046303   0.055309   0.837  0.402753
## SILVER.l1     0.029177   0.033716   0.865  0.387099
## COPPER.l1     0.055505   0.030267   1.834  0.067060 .
## ALUMINUM.l1  0.141546   0.039961   3.542  0.000421 ***
## GOLD.l2      -0.030449   0.055124  -0.552  0.580852
## SILVER.l2     0.036090   0.033717   1.070  0.284777
## COPPER.l2     0.033355   0.030354   1.099  0.272172
## ALUMINUM.l2  -0.004675   0.039992  -0.117  0.906964
## const        0.001084   0.001683   0.644  0.519574
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

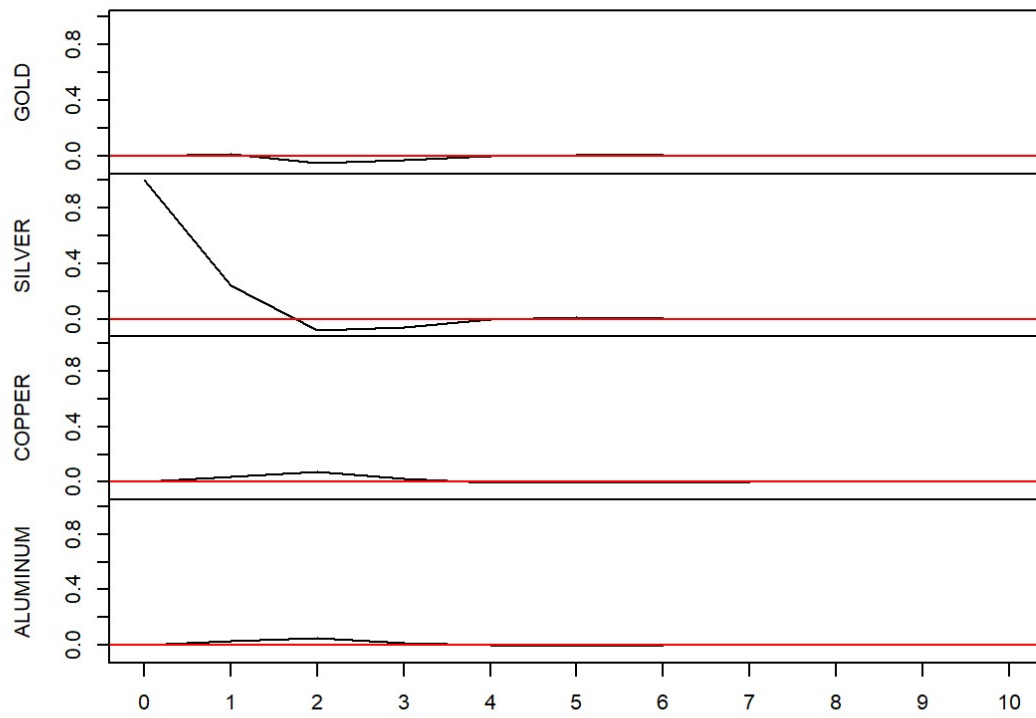
```

```
##
##
## Residual standard error: 0.04634 on 774 degrees of freedom
## Multiple R-Squared: 0.06348, Adjusted R-squared: 0.0538
## F-statistic: 6.558 on 8 and 774 DF, p-value: 2.673e-08
##
##
##
## Covariance matrix of residuals:
##          GOLD    SILVER    COPPER    ALUMINUM
## GOLD      0.0017432 0.0020245 0.0006366 0.0003668
## SILVER     0.0020245 0.0048989 0.0012838 0.0008549
## COPPER     0.0006366 0.0012838 0.0038091 0.0012166
## ALUMINUM   0.0003668 0.0008549 0.0012166 0.0021470
##
## Correlation matrix of residuals:
##          GOLD SILVER COPPER ALUMINUM
## GOLD      1.0000 0.6928 0.2470  0.1896
## SILVER     0.6928 1.0000 0.2972  0.2636
## COPPER     0.2470 0.2972 1.0000  0.4254
## ALUMINUM   0.1896 0.2636 0.4254  1.0000
```

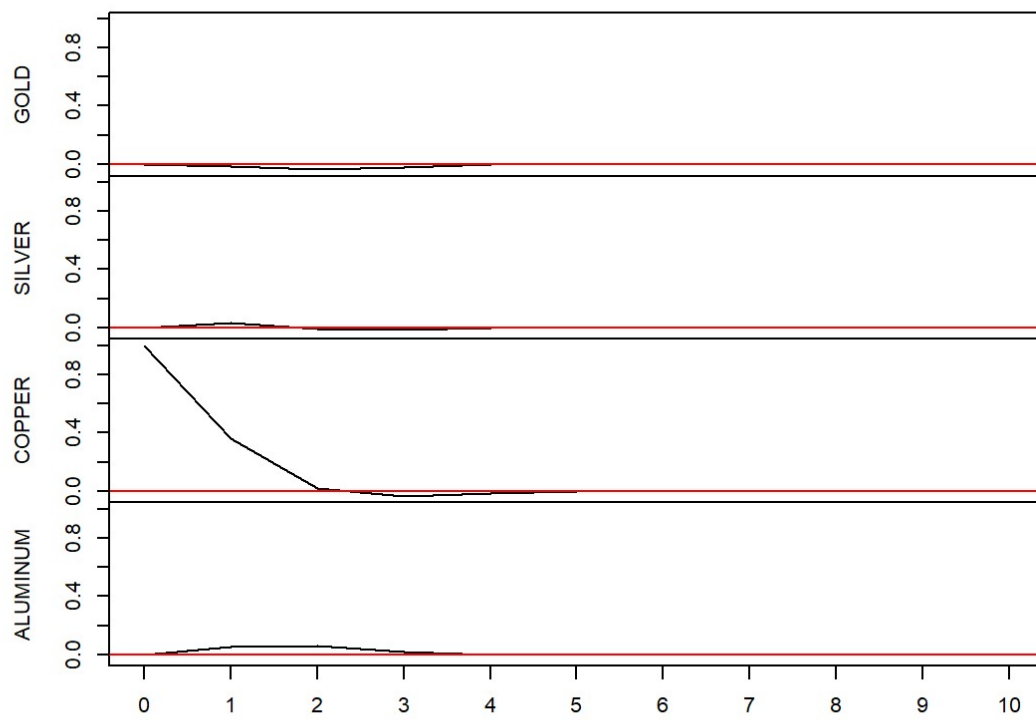
Impulse Response from GOLD



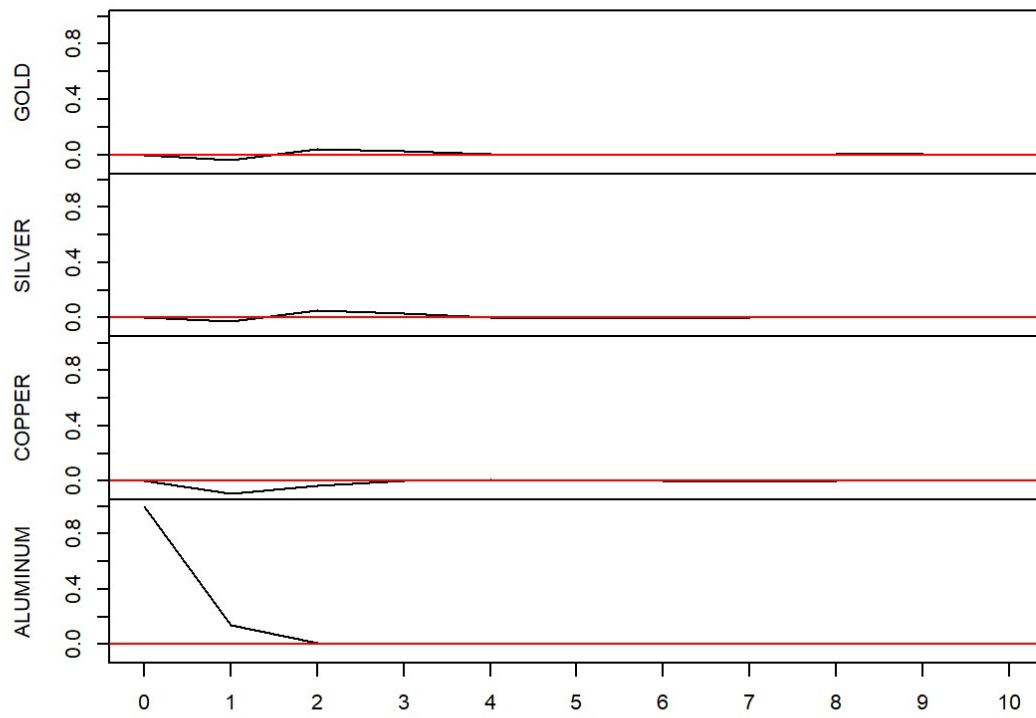
Impulse Response from SILVER



Impulse Response from COPPER



Impulse Response from ALUMINUM




```

##
## Conducting Johansen Cointegration Test:
## Johansen cointegration test summary:
##
## #####
## # Johansen-Procedure #
## #####
##
## Test type: trace statistic , without linear trend and constant in cointegration
##
## Eigenvalues (lambda):
## [1] 3.043522e-02 1.693186e-02 1.313910e-02 7.504290e-03 7.720265e-18
##
## Values of teststatistic and critical values of test:
##
##          test 10pct  5pct  1pct
## r <= 3 |  5.90   7.52   9.24 12.97
## r <= 2 | 16.25  17.85  19.96 24.60
## r <= 1 | 29.63  32.00  34.91 41.07
## r = 0  | 53.83  49.65  53.12 60.16
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##          GOLD.13  SILVER.13  COPPER.13  ALUMINUM.13  constant
## GOLD.13      1.0000000  1.0000000  1.0000000  1.0000000  1.0000000
## SILVER.13    -0.8128441  17.468770  -3.370131  -1.231297  -0.4704932
## COPPER.13    -0.1237594  -7.248443   4.131058  -1.818874   0.1998483
## ALUMINUM.13  -0.9670723 -26.926232  -2.675835   2.169833  -0.3813495
## constant      3.5036476 215.192062 -11.573299  -4.134316  -3.6196985
##
## Weights W:
## (This is the loading matrix)
##
##          GOLD.13  SILVER.13  COPPER.13  ALUMINUM.13  constant
## GOLD.d      -0.007135961 2.060427e-04  0.0019178214  0.0013594416  2.265961e-18
## SILVER.d      0.014469671 1.190717e-05  0.0034499982  0.0019397103  4.763093e-17
## COPPER.d      0.011358374 4.399753e-04 -0.0005692283  0.0037667659 -2.177216e-17
## ALUMINUM.d    0.011007700 6.205720e-04  0.0004071199 -0.0004845905 -9.253970e-17
##
## Estimated cointegration rank (5% significance): 1
##
## Fitting VECM Model:
## VECM fit summary:
## #####
## ###Model VECM
## #####
## Full sample size: 786      End sample size: 783
## Number of variables: 4      Number of estimated slope parameters 40
## AIC -19046.74      BIC -18846.22      SSR 9.684853
## Cointegrating vector (estimated by ML):
##      GOLD      SILVER      COPPER  ALUMINUM
## r1      1 -0.7724277 -0.1109516 -1.077521
##

```

```
##
##          ECT          Intercept          GOLD -1
## Equation GOLD      -0.0024(0.0047)      -0.0056(0.0200)      0.2969(0.0499)***
## Equation SILVER     0.0200(0.0079)*      0.0882(0.0335)**      0.0796(0.0833)
## Equation COPPER      0.0151(0.0070)*      0.0663(0.0295)*      0.0615(0.0735)
## Equation ALUMINUM    0.0145(0.0052)**      0.0625(0.0221)**      0.0498(0.0551)
##          SILVER -1          COPPER -1          ALUMINUM -1
## Equation GOLD      0.0090(0.0304)      -0.0115(0.0273)      -0.0413(0.0362)
## Equation SILVER     0.2416(0.0508)***      0.0341(0.0456)      -0.0089(0.0604)
## Equation COPPER      0.0413(0.0448)      0.3614(0.0402)***      -0.0830(0.0533)
## Equation ALUMINUM    0.0285(0.0336)      0.0540(0.0301).      0.1514(0.0399)***
##          GOLD -2          SILVER -2          COPPER -2
## Equation GOLD      -0.0459(0.0497)      -0.0590(0.0304).      -0.0220(0.0274)
## Equation SILVER     0.0042(0.0830)      -0.1405(0.0508)**      -0.0218(0.0457)
## Equation COPPER      -0.0616(0.0733)      0.0568(0.0448)      -0.1077(0.0403)**
## Equation ALUMINUM    -0.0329(0.0549)      0.0397(0.0336)      0.0346(0.0302)
##          ALUMINUM -2
## Equation GOLD      0.0562(0.0363)
## Equation SILVER     0.0901(0.0607)
## Equation COPPER      0.0317(0.0536)
## Equation ALUMINUM    0.0092(0.0401)
```

```
do_analysis(data_raw, "Wheat", wheat_cols)
```

```
##
## =====
## Analyzing Wheat Commodities
## =====
## Rows before cleaning: 786. After omitting NA: 551. Removed: 235
##
## ADF Test for Stationarity:
## ADF Test Results (on differenced log prices):
```

```
## WHEAT_US_HRW      : ADF Stat = -7.875, p-value = 0.0100 --> Stationary
```

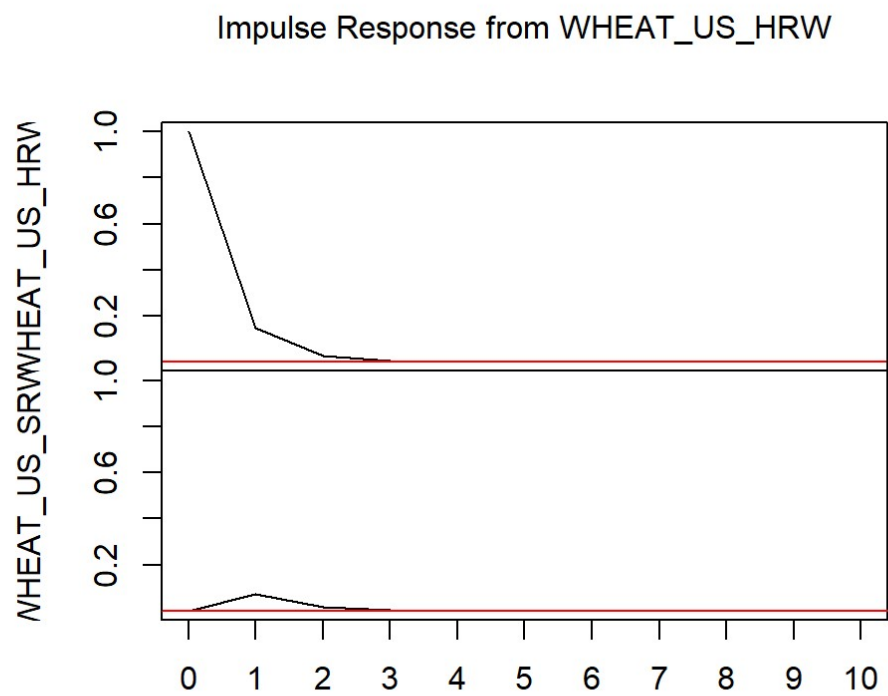
```
## WHEAT_US_SRW      : ADF Stat = -8.812, p-value = 0.0100 --> Stationary
```

```

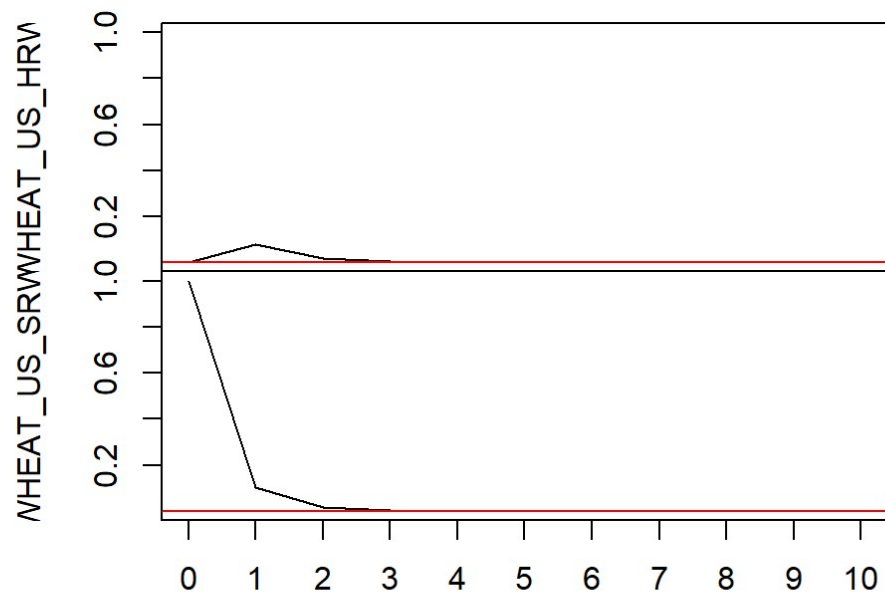
## Fitting VAR Model:
## Lag order selection (AIC, BIC, HQIC):
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      1      1      1      1
## Selected lag order by AIC: 1
## VAR Model summary:
## [1] "VAR Model object created. Checking its structure:"
##
## VAR Estimation Results:
## =====
## Endogenous variables: WHEAT_US_HRW, WHEAT_US_SRW
## Deterministic variables: const
## Sample size: 549
## Log Likelihood: 1808.534
## Roots of the characteristic polynomial:
## 0.2061 0.04534
## Call:
## VAR(y = diff_df, p = selected_lag, type = "const")
##
##
## Estimation results for equation WHEAT_US_HRW:
## =====
## WHEAT_US_HRW = WHEAT_US_HRW.l1 + WHEAT_US_SRW.l1 + const
##
##              Estimate Std. Error t value Pr(>|t|)
## WHEAT_US_HRW.l1 0.1476436  0.0705840   2.092  0.0369 *
## WHEAT_US_SRW.l1 0.0800912  0.0618441   1.295  0.1959
## const           0.0007652  0.0023996   0.319  0.7499
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.05622 on 546 degrees of freedom
## Multiple R-Squared: 0.05195, Adjusted R-squared: 0.04847
## F-statistic: 14.96 on 2 and 546 DF,  p-value: 4.736e-07
##
##
## Estimation results for equation WHEAT_US_SRW:
## =====
## WHEAT_US_SRW = WHEAT_US_HRW.l1 + WHEAT_US_SRW.l1 + const
##
##              Estimate Std. Error t value Pr(>|t|)
## WHEAT_US_HRW.l1 0.074680  0.081647   0.915  0.361
## WHEAT_US_SRW.l1 0.103813  0.071537   1.451  0.147
## const           0.000580  0.002776   0.209  0.835
##
##
## Residual standard error: 0.06503 on 546 degrees of freedom
## Multiple R-Squared: 0.02603, Adjusted R-squared: 0.02246
## F-statistic: 7.296 on 2 and 546 DF,  p-value: 0.0007463
##
##
##
## Covariance matrix of residuals:

```

```
##           WHEAT_US_HRW WHEAT_US_SRW
## WHEAT_US_HRW      0.003160      0.002931
## WHEAT_US_SRW      0.002931      0.004228
##
## Correlation matrix of residuals:
##           WHEAT_US_HRW WHEAT_US_SRW
## WHEAT_US_HRW      1.0000      0.8019
## WHEAT_US_SRW      0.8019      1.0000
```



Impulse Response from WHEAT_US_SRW



```

##
## Conducting Johansen Cointegration Test:
## Johansen cointegration test summary:
##
## #####
## # Johansen-Procedure #
## #####
##
## Test type: trace statistic , without linear trend and constant in cointegration
##
## Eigenvalues (lambda):
## [1] 6.383987e-02 1.204309e-02 -8.660055e-20
##
## Values of teststatistic and critical values of test:
##
##          test 10pct  5pct  1pct
## r <= 1 | 6.65  7.52  9.24 12.97
## r = 0  | 42.87 17.85 19.96 24.60
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##          WHEAT_US_HRW.l2 WHEAT_US_SRW.l2  constant
## WHEAT_US_HRW.l2      1.0000000      1.0000000  1.0000000
## WHEAT_US_SRW.l2     -1.0578153      0.1436331 -0.5649347
## constant           0.2111572     -5.9759910 -0.7656697
##
## Weights W:
## (This is the loading matrix)
##
##          WHEAT_US_HRW.l2 WHEAT_US_SRW.l2  constant
## WHEAT_US_HRW.d     -0.06304098     -0.01453450  8.109859e-17
## WHEAT_US_SRW.d      0.04297320     -0.01779454 -2.149884e-17
##
## Estimated cointegration rank (5% significance): 1
##
## Fitting VECM Model:
## VECM fit summary:
## #####
## ###Model VECM
## #####
## Full sample size: 551    End sample size: 549
## Number of variables: 2   Number of estimated slope parameters 8
## AIC -6751.235    BIC -6712.462    SSR 4.009082
## Cointegrating vector (estimated by ML):
##      WHEAT_US_HRW WHEAT_US_SRW
## r1          1      -1.058019
##
##
##          ECT          Intercept          WHEAT_US_HRW -1
## Equation WHEAT_US_HRW -0.0628(0.0270)* -0.0127(0.0063)* 0.1670(0.0708)*
## Equation WHEAT_US_SRW 0.0433(0.0313) 0.0099(0.0073) 0.0613(0.0821)
##          WHEAT_US_SRW -1
## Equation WHEAT_US_HRW 0.0465(0.0633)

```

```
## Equation WHEAT_US_SRW 0.1270(0.0734).
```

```
do_analysis(data_raw, "Sugar", sugar_cols)
```

```
##  
## =====  
## Analyzing Sugar Commodities  
## =====  
## Rows before cleaning: 786. After omitting NA: 786. Removed: 0  
##  
## ADF Test for Stationarity:  
## ADF Test Results (on differenced log prices):
```

```
## SUGAR_EU      : ADF Stat = -8.806, p-value = 0.0100 --> Stationary
```

```
## SUGAR_US      : ADF Stat = -8.750, p-value = 0.0100 --> Stationary
```

```
## SUGAR_WLD     : ADF Stat = -8.297, p-value = 0.0100 --> Stationary
```

```

## Fitting VAR Model:
## Lag order selection (AIC, BIC, HQIC):
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      7      2      1      7
## Selected lag order by AIC: 7
## VAR Model summary:
## [1] "VAR Model object created. Checking its structure:"
##
## VAR Estimation Results:
## =====
## Endogenous variables: SUGAR_EU, SUGAR_US, SUGAR_WLD
## Deterministic variables: const
## Sample size: 778
## Log Likelihood: 3385.669
## Roots of the characteristic polynomial:
## 0.8358 0.8358 0.8137 0.8137 0.8004 0.7664 0.7664 0.7625 0.7625 0.6614 0.6614 0.6547 0.6
547 0.6421 0.6421 0.6268 0.5882 0.5882 0.5436 0.4465 0.05488
## Call:
## VAR(y = diff_df, p = selected_lag, type = "const")
##
##
## Estimation results for equation SUGAR_EU:
## =====
## SUGAR_EU = SUGAR_EU.l1 + SUGAR_US.l1 + SUGAR_WLD.l1 + SUGAR_EU.l2 + SUGAR_US.l2 + SUGAR
_WLD.l2 + SUGAR_EU.l3 + SUGAR_US.l3 + SUGAR_WLD.l3 + SUGAR_EU.l4 + SUGAR_US.l4 + SUGAR_WLD
.l4 + SUGAR_EU.l5 + SUGAR_US.l5 + SUGAR_WLD.l5 + SUGAR_EU.l6 + SUGAR_US.l6 + SUGAR_WLD.l6
+ SUGAR_EU.l7 + SUGAR_US.l7 + SUGAR_WLD.l7 + const
##
##
##           Estimate Std. Error t value Pr(>|t|)
## SUGAR_EU.l1 -0.166255    0.036297  -4.580 5.43e-06 ***
## SUGAR_US.l1 -0.057012    0.037236  -1.531 0.126162
## SUGAR_WLD.l1  0.038471    0.020603   1.867 0.062255 .
## SUGAR_EU.l2 -0.028738    0.037302  -0.770 0.441292
## SUGAR_US.l2  0.192789    0.037983   5.076 4.87e-07 ***
## SUGAR_WLD.l2  0.001625    0.021370   0.076 0.939412
## SUGAR_EU.l3 -0.015822    0.037164  -0.426 0.670418
## SUGAR_US.l3 -0.049635    0.038197  -1.299 0.194183
## SUGAR_WLD.l3 -0.023059    0.021399  -1.078 0.281560
## SUGAR_EU.l4  0.154991    0.036618   4.233 2.59e-05 ***
## SUGAR_US.l4  0.012075    0.038238   0.316 0.752260
## SUGAR_WLD.l4  0.011826    0.021452   0.551 0.581605
## SUGAR_EU.l5 -0.164812    0.037056  -4.448 9.99e-06 ***
## SUGAR_US.l5  0.054952    0.038026   1.445 0.148838
## SUGAR_WLD.l5 -0.004091    0.021471  -0.191 0.848928
## SUGAR_EU.l6 -0.038766    0.037138  -1.044 0.296900
## SUGAR_US.l6 -0.059106    0.038140  -1.550 0.121625
## SUGAR_WLD.l6 -0.002301    0.021450  -0.107 0.914619
## SUGAR_EU.l7 -0.004058    0.036240  -0.112 0.910880
## SUGAR_US.l7  0.130710    0.037028   3.530 0.000441 ***
## SUGAR_WLD.l7 -0.027403    0.020705  -1.324 0.186065
## const       0.001271    0.001652   0.769 0.441960
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```



```

##
## Residual standard error: 0.04579 on 756 degrees of freedom
## Multiple R-Squared: 0.16, Adjusted R-squared: 0.1367
## F-statistic: 6.859 on 21 and 756 DF, p-value: < 2.2e-16
##
##
## Estimation results for equation SUGAR_US:
## =====
## SUGAR_US = SUGAR_EU.11 + SUGAR_US.11 + SUGAR_WLD.11 + SUGAR_EU.12 + SUGAR_US.12 + SUGAR
_WLD.12 + SUGAR_EU.13 + SUGAR_US.13 + SUGAR_WLD.13 + SUGAR_EU.14 + SUGAR_US.14 + SUGAR_WLD
.14 + SUGAR_EU.15 + SUGAR_US.15 + SUGAR_WLD.15 + SUGAR_EU.16 + SUGAR_US.16 + SUGAR_WLD.16
+ SUGAR_EU.17 + SUGAR_US.17 + SUGAR_WLD.17 + const
##
##
## Estimate Std. Error t value Pr(>|t|)
## SUGAR_EU.11 0.178435 0.041009 4.351 1.54e-05 ***
## SUGAR_US.11 0.191611 0.042070 4.555 6.12e-06 ***
## SUGAR_WLD.11 0.025526 0.023278 1.097 0.273161
## SUGAR_EU.12 0.108057 0.042144 2.564 0.010540 *
## SUGAR_US.12 -0.051948 0.042915 -1.211 0.226461
## SUGAR_WLD.12 -0.024204 0.024144 -1.002 0.316429
## SUGAR_EU.13 -0.028107 0.041988 -0.669 0.503440
## SUGAR_US.13 0.055352 0.043155 1.283 0.200016
## SUGAR_WLD.13 0.037276 0.024177 1.542 0.123537
## SUGAR_EU.14 -0.114516 0.041372 -2.768 0.005779 **
## SUGAR_US.14 -0.014668 0.043202 -0.340 0.734314
## SUGAR_WLD.14 0.027368 0.024237 1.129 0.259183
## SUGAR_EU.15 -0.157165 0.041867 -3.754 0.000187 ***
## SUGAR_US.15 0.124930 0.042963 2.908 0.003746 **
## SUGAR_WLD.15 0.010526 0.024258 0.434 0.664481
## SUGAR_EU.16 0.021389 0.041960 0.510 0.610366
## SUGAR_US.16 -0.027292 0.043091 -0.633 0.526693
## SUGAR_WLD.16 0.020642 0.024235 0.852 0.394626
## SUGAR_EU.17 0.116352 0.040944 2.842 0.004608 **
## SUGAR_US.17 -0.121159 0.041835 -2.896 0.003887 **
## SUGAR_WLD.17 -0.014351 0.023393 -0.613 0.539748
## const 0.001586 0.001866 0.849 0.395883
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.05173 on 756 degrees of freedom
## Multiple R-Squared: 0.1624, Adjusted R-squared: 0.1391
## F-statistic: 6.978 on 21 and 756 DF, p-value: < 2.2e-16
##
##
## Estimation results for equation SUGAR_WLD:
## =====
## SUGAR_WLD = SUGAR_EU.11 + SUGAR_US.11 + SUGAR_WLD.11 + SUGAR_EU.12 + SUGAR_US.12 + SUGA
R_WLD.12 + SUGAR_EU.13 + SUGAR_US.13 + SUGAR_WLD.13 + SUGAR_EU.14 + SUGAR_US.14 + SUGAR_WL
D.14 + SUGAR_EU.15 + SUGAR_US.15 + SUGAR_WLD.15 + SUGAR_EU.16 + SUGAR_US.16 + SUGAR_WLD.16
+ SUGAR_EU.17 + SUGAR_US.17 + SUGAR_WLD.17 + const
##
##
## Estimate Std. Error t value Pr(>|t|)
## SUGAR_EU.11 0.175850 0.075031 2.344 0.0194 *
## SUGAR_US.11 0.004217 0.076972 0.055 0.9563

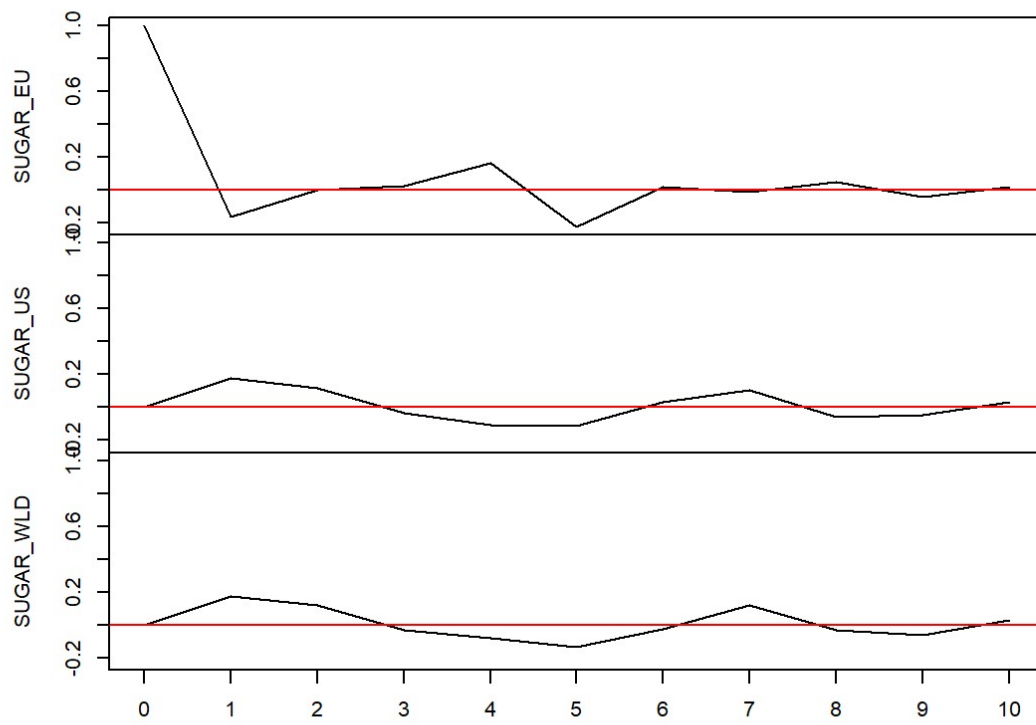
```

```

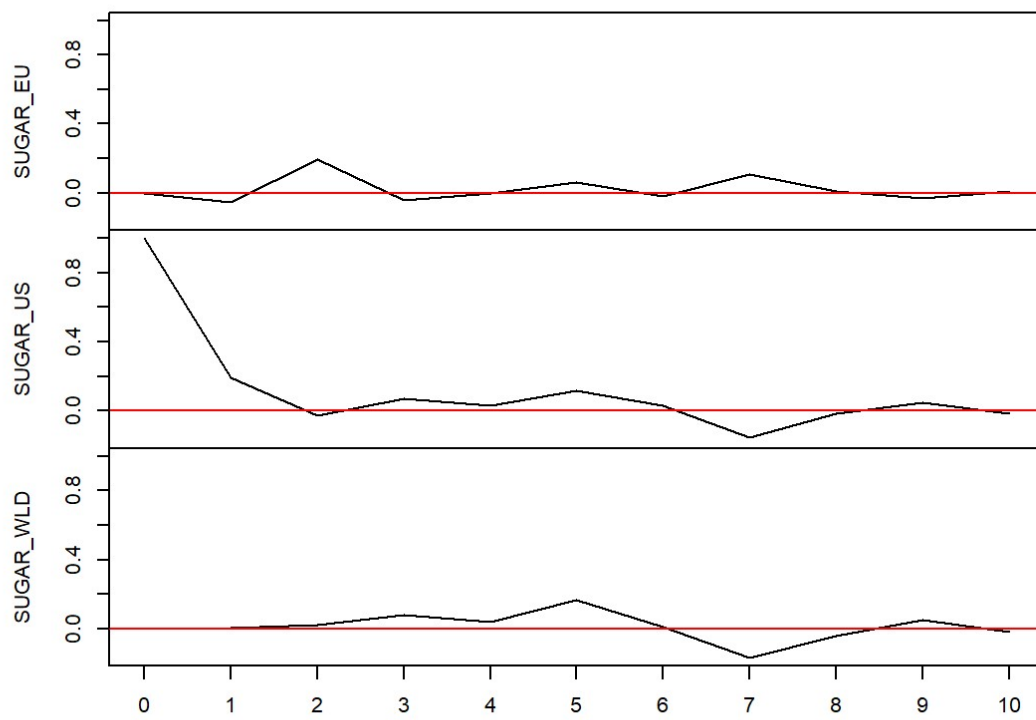
## SUGAR_WLD.11  0.302967  0.042589  7.114 2.62e-12 ***
## SUGAR_EU.12   0.098069  0.077108  1.272  0.2038
## SUGAR_US.12   0.030232  0.078517  0.385  0.7003
## SUGAR_WLD.12 -0.114003  0.044175 -2.581  0.0100 *
## SUGAR_EU.13   -0.038139  0.076823 -0.496  0.6197
## SUGAR_US.13   0.039135  0.078958  0.496  0.6203
## SUGAR_WLD.13  0.045717  0.044234  1.034  0.3017
## SUGAR_EU.14   -0.080645  0.075695 -1.065  0.2870
## SUGAR_US.14   -0.004678  0.079043 -0.059  0.9528
## SUGAR_WLD.14  0.029806  0.044344  0.672  0.5017
## SUGAR_EU.15   -0.173604  0.076600 -2.266  0.0237 *
## SUGAR_US.15   0.169758  0.078605  2.160  0.0311 *
## SUGAR_WLD.15 -0.064394  0.044384 -1.451  0.1472
## SUGAR_EU.16   -0.014861  0.076770 -0.194  0.8466
## SUGAR_US.16   -0.085112  0.078840 -1.080  0.2807
## SUGAR_WLD.16  0.075317  0.044341  1.699  0.0898 .
## SUGAR_EU.17   0.139866  0.074913  1.867  0.0623 .
## SUGAR_US.17   -0.114762  0.076542 -1.499  0.1342
## SUGAR_WLD.17 -0.022145  0.042800 -0.517  0.6050
## const         0.001267  0.003415  0.371  0.7106
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.09465 on 756 degrees of freedom
## Multiple R-Squared: 0.1248, Adjusted R-squared: 0.1005
## F-statistic: 5.134 on 21 and 756 DF, p-value: 1.253e-12
##
##
## Covariance matrix of residuals:
##          SUGAR_EU  SUGAR_US  SUGAR_WLD
## SUGAR_EU  0.0020967 0.0002242 0.0005167
## SUGAR_US  0.0002242 0.0026764 0.0025266
## SUGAR_WLD 0.0005167 0.0025266 0.0089593
##
## Correlation matrix of residuals:
##          SUGAR_EU  SUGAR_US  SUGAR_WLD
## SUGAR_EU  1.00000  0.09463  0.1192
## SUGAR_US  0.09463  1.00000  0.5160
## SUGAR_WLD 0.11922  0.51596  1.0000

```

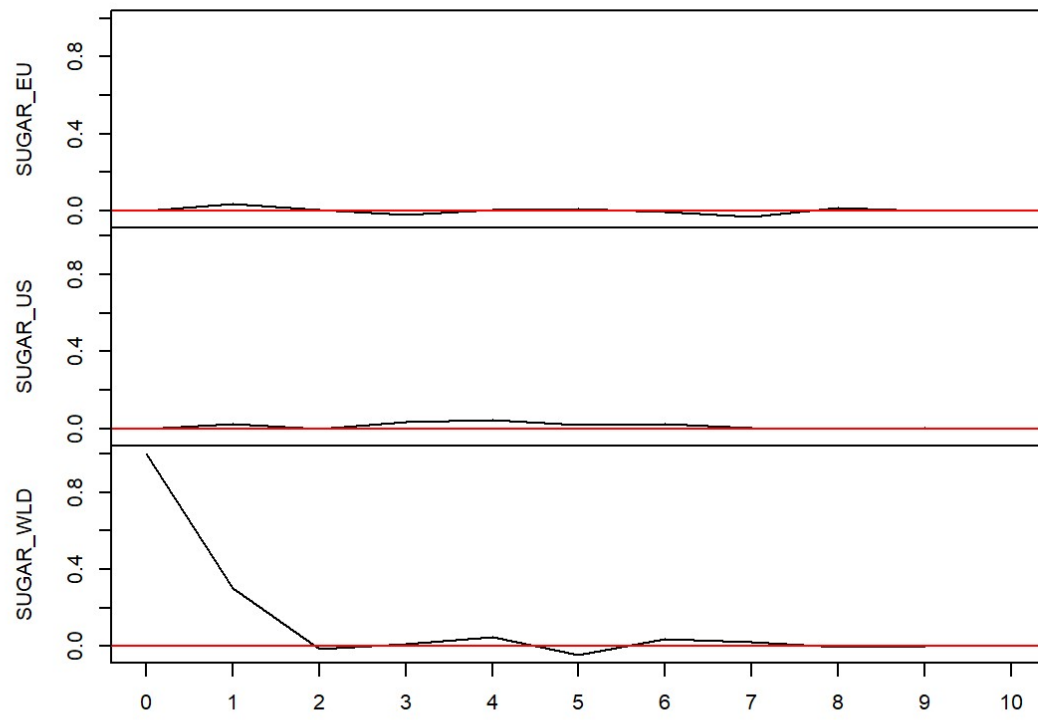
Impulse Response from SUGAR_EU



Impulse Response from SUGAR_US



Impulse Response from SUGAR_WLD



```

##
## Conducting Johansen Cointegration Test:
## Johansen cointegration test summary:
##
## #####
## # Johansen-Procedure #
## #####
##
## Test type: trace statistic , without linear trend and constant in cointegration
##
## Eigenvalues (lambda):
## [1] 1.801699e-02 9.235970e-03 4.839240e-03 1.040834e-17
##
## Values of teststatistic and critical values of test:
##
##          test 10pct  5pct  1pct
## r <= 2 |   3.77   7.52   9.24 12.97
## r <= 1 |  10.99  17.85  19.96 24.60
## r = 0  |  25.14  32.00  34.91 41.07
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##          SUGAR_EU.18 SUGAR_US.18 SUGAR_WLD.18  constant
## SUGAR_EU.18    1.000000    1.000000    1.000000    1.000000
## SUGAR_US.18   -6.297247   -1.7737121  -0.66794891  0.9831396
## SUGAR_WLD.18    4.964015   -0.1045832    0.08188919 -0.2672651
## constant        3.323234   -0.5730108    0.40191220  3.2205703
##
## Weights W:
## (This is the loading matrix)
##
##          SUGAR_EU.18 SUGAR_US.18 SUGAR_WLD.18  constant
## SUGAR_EU.d    0.0002678957 0.0002412534 -0.0078054787 2.288141e-18
## SUGAR_US.d    0.0006721669 0.0070969571 -0.0003904985 2.227666e-18
## SUGAR_WLD.d -0.0051010496 0.0086212234 -0.0025127696 5.599389e-18
##
## Estimated cointegration rank (5% significance): 0
##
## No significant cointegration detected; skipping VECM.

```